

CompTIA A+

Core 1 (220-1101) + Core 2 (220-1102)

300 Interview Questions & Answers

Certification	CompTIA A+
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How to use this guide: Each question (Q) is followed by a detailed answer. Green TIP boxes highlight exam-critical notes. Core 1 domains: Mobile Devices, Networking, Hardware, Virtualization/Cloud, Hardware Troubleshooting. Core 2 domains: Operating Systems, Security, Software Troubleshooting, Operational Procedures.

SECTION 01

Mobile Devices

Laptops, Smartphones, Tablets, Peripherals, Accessories

Q1 – Q30

Q1 What are the main components inside a laptop that differ from a desktop?

Ans: Laptops use smaller, power-efficient components: SO-DIMM RAM (smaller than DIMM), M.2 or 2.5-inch SSD/HDD, integrated battery, built-in display/keyboard/touchpad, and soldered CPU in ultrabooks. They use integrated or MXM graphics instead of full-size GPUs.

TIP *Core 1 Exam Domain 1 – Mobile Devices is ~15% of the exam.*

Q2 What is SO-DIMM and how does it differ from DIMM?

Ans: SO-DIMM (Small Outline DIMM) is a smaller RAM module used in laptops, mini PCs, and printers. It has 260 pins (DDR4) vs 288 pins on desktop DIMM. Functionally identical in performance; just physically smaller to fit compact form factors.

Q3 What are the types of laptop storage form factors?

Ans: Laptop storage types: 2.5-inch SATA SSD/HDD (7mm or 9.5mm thick), M.2 SATA (slower), M.2 NVMe PCIe (fastest, 3500+ MB/s), and mSATA (older, rare). M.2 NVMe is now standard in modern laptops for speed and compact size.

Q4 What is a docking station?

Ans: A docking station expands laptop connectivity by providing additional ports: USB-A/C, HDMI, DisplayPort, Ethernet, SD card, and audio jacks — all through a single cable (USB-C/Thunderbolt) connection. Enables desktop-like setup when docking a laptop at a workstation.

Q5 What is Thunderbolt and how does it differ from USB-C?

Ans: Thunderbolt (currently v4/5) uses the USB-C physical connector but delivers significantly higher bandwidth (40-120 Gbps), daisy-chaining of up to 6 devices, power delivery, and display output. All Thunderbolt ports are USB-C, but not all USB-C ports support Thunderbolt.

Q6 How do you upgrade RAM in a laptop?

Ans: Check laptop service manual for accessible SO-DIMM slots (some have soldered RAM). Power off, remove battery/disconnect AC, open bottom panel (Phillips screws), locate RAM slots, release retention clips, remove old module at 45 degrees, insert new module at 45 degrees until clips click, reassemble and verify in BIOS/Task Manager.

Q7 What is a laptop inverter and when does it fail?

Ans: The inverter converts DC power to AC for CCFL (older fluorescent) backlights. Symptoms of inverter failure: dim/flickering display, pink tint, or no backlight (screen visible with flashlight). LED backlit displays do not use inverters. Inverter failure requires replacement of the inverter board.

Q8 What is the difference between IPS, TN, and OLED laptop displays?

Ans: TN (Twisted Nematic): fast response, low cost, poor viewing angles and color — budget gaming laptops. IPS (In-Plane Switching): excellent color accuracy and wide viewing angles — professional and mainstream laptops. OLED: perfect blacks, vivid color, thin — premium ultrabooks; risk of burn-in.

Q9 What is a digitizer on a mobile device?

Ans: A digitizer is a transparent layer over the display that detects stylus or finger touch input. It sits between the glass and LCD panel. When a screen cracks but touch still works, the glass is broken; if touch fails, the digitizer is damaged. Replacement requires careful disassembly.

Q10 What wireless technologies are found in mobile devices?

Ans: Mobile wireless technologies: Wi-Fi (802.11ax/Wi-Fi 6), Bluetooth (5.x), NFC (contactless payments/pairing), GPS (positioning), cellular (4G LTE/5G), IR blaster (remote control), and Ultra-Wideband (UWB) for precise spatial positioning (AirDrop, car keys).

Q11 What is the difference between CDMA and GSM cellular technologies?

Ans: GSM (Global System for Mobile) uses SIM cards for subscriber identity — common worldwide; carriers include AT&T, T-Mobile. CDMA (Code Division Multiple Access) encodes calls across a frequency band — used by Verizon (US), Sprint. Modern 4G LTE and 5G networks use SIM on all carriers, making CDMA largely obsolete.

Q12 What is airplane mode?

Ans: Airplane mode disables all wireless transmissions: cellular, Wi-Fi, Bluetooth, NFC, and GPS. Required during flights to prevent interference with aircraft navigation. Individual radios (Wi-Fi, Bluetooth) can often be re-enabled individually after activating airplane mode.

Q13 How do you extend laptop battery life?

Ans: Reduce screen brightness (biggest power consumer), enable battery saver mode, disable unused radios (Wi-Fi, Bluetooth), close background apps, lower CPU performance mode, use SSD instead of HDD, avoid extreme temperatures, and calibrate battery periodically.

Q14 What is a laptop CPU vs a desktop CPU?

Ans: Laptop CPUs (labeled 'U', 'H', 'HX' in Intel; 'U', 'HS', 'HX' in AMD) prioritize power efficiency and thermal design over raw performance. TDP ranges: 9-28W (U-series) vs 45-55W (H-series) vs 125W+ (desktop). Laptop CPUs are usually soldered (BGA) and cannot be upgraded.

Q15 What is the function of a laptop touchpad?

Ans: A touchpad (trackpad) is a touch-sensitive input device replacing the mouse on laptops. Modern precision touchpads support multi-touch gestures: two-finger scroll, pinch-to-zoom, three-finger swipe. They use capacitive sensing. Issues: sensitivity, driver problems, or physical damage.

Q16 What are the common laptop screen resolutions?

Ans: Common laptop resolutions: 1366x768 (HD, budget), 1920x1080 (FHD, most common), 2560x1440 (QHD, premium), 3840x2160 (4K UHD, high-end), 2880x1800 (Retina MacBook). Higher resolution improves clarity but increases GPU load and reduces battery life.

Q17 What is a KVM extender in mobile context?

Ans: In mobile/remote contexts, KVM-over-IP allows IT admins to remotely access servers and computers at a hardware level (BIOS access, OS boot) using a network connection. Different from software remote desktop — works even when OS is not running.

Q18 What is an e-reader and how does it display content?

Ans: An e-reader (like Kindle) uses E-Ink technology — electronic paper that reflects ambient light like real paper, unlike backlit LCD/OLED displays. Benefits: no eye strain, weeks of battery life, readable in sunlight. Limitations: slow refresh rate, limited color (mostly grayscale), no video.

Q19 What is a mobile hotspot?

Ans: A mobile hotspot shares a smartphone or cellular device's internet connection via Wi-Fi to other devices. The phone acts as a wireless router using cellular data. Can also be a dedicated MiFi device. Usage consumes cellular data and battery faster.

Q20 What is the difference between Android and iOS from an IT support perspective?

Ans: Android: open platform, varies by manufacturer, root access possible, sideloading apps allowed, more fragmentation in updates. iOS: closed ecosystem, uniform hardware, no sideloading (without jailbreak), consistent updates for 5-6 years, more secure by design but less flexible.

Q21 What is MDM (Mobile Device Management)?

Ans: MDM is software that allows IT administrators to remotely manage, monitor, configure, and secure mobile devices (smartphones, tablets, laptops) in an organization. Features: remote wipe, policy enforcement, app management, compliance monitoring. Examples: Microsoft Intune, VMware Workspace ONE, Jamf.

TIP *Key enterprise IT skill – MDM appears on CompTIA A+ Core 2 exam.*

Q22 What is remote wipe on a mobile device?

Ans: Remote wipe erases all data on a lost or stolen mobile device remotely via MDM or built-in services (Find My iPhone, Android Device Manager). Factory resets the device. Essential for protecting sensitive corporate data. Ensure data is backed up regularly.

Q23 What is screen mirroring?

Ans: Screen mirroring wirelessly transmits a device's display to a TV or monitor. Technologies: AirPlay (Apple), Miracast (cross-platform), Google Cast (Chromecast). Useful for presentations and media consumption. Requires compatible hardware on both devices.

Q24 What is the difference between MicroSD and SD cards?

Ans: MicroSD is a smaller form of the SD card standard used in smartphones, tablets, and cameras. Sizes: Standard SD (32x24mm), MiniSD (21x20mm, rare), MicroSD (15x11mm). Adapter allows microSD in standard SD slots. Speed classes (C10, U1/U3, V30, A1/A2) indicate write speeds.

Q25 What is a biometric authentication method on mobile devices?

Ans: Biometric authentication uses unique physical characteristics: fingerprint scanner (capacitive sensor in screen or button), facial recognition (2D camera or 3D structured light like Face ID), iris scanner. More convenient than PINs; security depends on implementation quality.

Q26 What is the purpose of a TPM chip in a laptop?

Ans: TPM (Trusted Platform Module) is a dedicated security chip that stores cryptographic keys, certificates, and passwords in hardware. Used for: BitLocker drive encryption key storage, secure boot verification, platform integrity measurement. Required for Windows 11 installation.

Q27 What are common laptop hardware failures and symptoms?

Ans: Common failures: No power (bad battery, AC adapter, DC jack), no display (GPU, inverter, cable), keyboard failure (spill damage, worn keys), overheating (fan failure, thermal paste), Wi-Fi issues (driver or antenna), battery not charging (battery or charging circuit).

Q28 What is a laptop AC adapter and how to identify compatibility?

Ans: AC adapter converts mains AC to DC power for the laptop. Key specs: voltage (V), current (A), wattage ($W=V \times A$), and connector tip size/polarity. Using wrong voltage/polarity can damage the laptop. Modern USB-C PD (Power Delivery) adapters are universal up to 100-240W.

Q29 What is the difference between a 2-in-1 laptop and a tablet?

Ans: 2-in-1 laptops (convertible/hybrid) combine a full laptop keyboard with a touchscreen that can fold 360 degrees (convertible) or detach (detachable). They run desktop OS (Windows). Tablets run mobile OS (iOS, Android) and are primarily touch-driven without a physical keyboard.

Q30 What is DisplayPort over USB-C / Alt Mode?

Ans: Alt Mode (Alternate Mode) allows USB-C ports to carry non-USB signals. DisplayPort Alt Mode lets a USB-C port output video directly to a DisplayPort monitor. Thunderbolt also supports this. Enables single-cable docking setups with video, data, and power delivery.

SECTION 02

Networking Fundamentals

Protocols, IP Addressing, Wireless, Network Hardware, Troubleshooting

Q31 – Q60

Q31 What is the OSI model and why is it important for A+ technicians?

Ans: The OSI (Open Systems Interconnection) model is a 7-layer framework standardizing network communication. Layers: Physical, Data Link, Network, Transport, Session, Presentation, Application. A+ technicians use it to isolate problems: Layer 1 issues are cable/NIC; Layer 3 issues are IP/routing; Layer 7 issues are application/browser.

TIP *Mnemonic for OSI layers bottom-up: 'Please Do Not Throw Sausage Pizza Away'*

Q32 What is TCP/IP and what are its 4 layers?

Ans: TCP/IP is the foundational internet protocol suite with 4 layers: Network Access (frames, MAC addresses – OSI L1+L2), Internet (IP routing – OSI L3), Transport (TCP/UDP – OSI L4), Application (HTTP, DNS, SMTP – OSI L5-7). All internet communication uses TCP/IP.

Q33 What is the difference between TCP and UDP?

Ans: TCP (Transmission Control Protocol): connection-oriented, three-way handshake, guaranteed delivery, ordered packets, error correction. Used for: HTTP/S, email, FTP. UDP (User Datagram Protocol): connectionless, no guarantee, low overhead, fast. Used for: DNS, VoIP, video streaming, online gaming.

Q34 What is an IP address and what are its classes?

Ans: IPv4 is a 32-bit address in dotted-decimal notation. Classes: A (1.0.0.0–126.x.x.x, /8), B (128.0.0.0–191.255.x.x, /16), C (192.0.0.0–223.255.255.x, /24). Private ranges: 10.x.x.x (A), 172.16–31.x.x (B), 192.168.x.x (C). 127.0.0.1 is loopback (localhost).

Q35 What is APIPA and when does it occur?

Ans: APIPA (Automatic Private IP Addressing) assigns addresses in the 169.254.x.x /16 range when a DHCP server is unreachable. It allows link-local communication only — no internet access. Seeing 169.254.x.x indicates DHCP failure: check DHCP server, network cable, or DHCP service.

TIP 169.254.x.x address is a dead giveaway of DHCP failure — exam favorite!

Q36 What is DHCP and how does DORA work?

Ans: DHCP automatically assigns IP configuration. DORA process: Discover (client broadcasts to find DHCP server), Offer (server offers an IP lease), Request (client requests the offered IP), Acknowledge (server confirms the lease). Default lease time varies; common is 24 hours or 8 days.

Q37 What is DNS and what record types exist?

Ans: DNS resolves domain names to IP addresses. Record types: A (IPv4 address), AAAA (IPv6 address), CNAME (alias/canonical name), MX (mail server), PTR (reverse lookup, IP to name), TXT (text, SPF/DKIM for email security), NS (name server), SOA (zone authority).

Q38 What are well-known port numbers a technician must memorize?

Ans: Critical ports: FTP 20/21, SSH 22, Telnet 23, SMTP 25, DNS 53, DHCP 67/68, HTTP 80, POP3 110, IMAP 143, HTTPS 443, SMB 445, RDP 3389, SNMP 161/162, LDAP 389, NTP 123, SFTP 22 (same as SSH).

TIP Port numbers are heavily tested on both Core 1 and Core 2 exams!

Q39 What is subnetting and what is a subnet mask?

Ans: Subnetting divides a large network into smaller subnets for efficiency and security. A subnet mask identifies network vs host portion. /24 = 255.255.255.0 (254 hosts), /25 = 255.255.255.128 (126 hosts). CIDR notation (/24) is shorthand for the subnet mask.

Q40 What is NAT and why is it used?

Ans: NAT (Network Address Translation) maps multiple private IP addresses to one or more public IPs. Types: SNAT (Static NAT, 1:1), DNAT (Dynamic NAT, pool), PAT/Overload (Port Address Translation, many:1). PAT allows thousands of devices to share one public IP — the most common home/office setup.

Q41 What is a VLAN and why are they used?

Ans: VLAN (Virtual LAN) logically segments a physical network into separate broadcast domains without requiring separate physical switches. Benefits: security (isolate guest Wi-Fi), performance (reduce broadcast traffic), management (group users by department). Configured on managed switches using IEEE 802.1Q tagging.

Q42 What is the difference between a hub, switch, and router?

Ans: Hub: Layer 1, broadcasts all data to all ports, creates single collision domain – obsolete. Switch: Layer 2, forwards frames based on MAC address to correct port, separate collision domains per port. Router: Layer 3, routes packets between different networks using IP addresses.

Q43 What is a managed vs unmanaged switch?

Ans: Unmanaged switches work out of the box with no configuration – plug and play for small offices. Managed switches offer VLAN support, QoS (Quality of Service), port mirroring, SNMP monitoring, spanning tree protocol, link aggregation, and remote management via CLI/Web GUI.

Q44 What is Wi-Fi 6 (802.11ax)?

Ans: Wi-Fi 6 (802.11ax) improves performance in dense environments using OFDMA (orthogonal frequency division multiple access) to serve multiple clients simultaneously, MU-MIMO (8x8), BSS Coloring to reduce interference, TWT (Target Wake Time) for IoT battery efficiency. Max theoretical: 9.6 Gbps.

Q45 What is the difference between 2.4 GHz and 5 GHz Wi-Fi bands?

Ans: 2.4 GHz: longer range, better wall penetration, more interference (microwaves, Bluetooth, neighboring networks), 3 non-overlapping channels (1, 6, 11). 5 GHz: shorter range, faster speeds, 25 non-overlapping channels, less interference. Dual-band routers offer both simultaneously.

Q46 What is WPA3 and how does it improve security over WPA2?

Ans: WPA3 (Wi-Fi Protected Access 3) improves: SAE (Simultaneous Authentication of Equals) replaces PSK handshake – protects against offline dictionary attacks; Forward Secrecy – compromised key doesn't decrypt past sessions; 192-bit encryption for enterprise; OWE (Opportunistic Wireless Encryption) for open networks.

Q47 What is a wireless access point (WAP) vs a wireless router?

Ans: A WAP provides wireless connectivity to an existing wired network — it bridges wireless clients to the LAN switch. It does not route between networks. A wireless router combines WAP + router + NAT + DHCP server in one device, designed for home/small office use connecting to an ISP.

Q48 What is PoE (Power over Ethernet)?

Ans: PoE delivers electrical power over Ethernet cables (Cat5e+) alongside data, eliminating separate power adapters. Used for: VoIP phones, WAPs, IP cameras, smart lighting. Standards: IEEE 802.3af (15.4W), 802.3at/PoE+ (30W), 802.3bt/PoE++ (60-100W). Requires PoE-capable switch or injector.

Q49 What is a proxy server?

Ans: A proxy server is an intermediary between clients and the internet. Benefits: caching (speeds up repeated requests), content filtering (block inappropriate sites), anonymity, and traffic logging. Transparent proxies work without client configuration; explicit proxies require browser settings.

Q50 What is QoS (Quality of Service)?

Ans: QoS prioritizes certain network traffic to ensure time-sensitive applications (VoIP, video conferencing) receive bandwidth and low latency even during congestion. Configured on routers and managed switches. Tags traffic with DSCP values. Essential in mixed office networks.

Q51 What is the difference between IPv4 and IPv6?

Ans: IPv4: 32-bit, ~4.3 billion addresses, dotted-decimal (192.168.1.1), requires NAT for address conservation. IPv6: 128-bit, 340 undecillion addresses, hexadecimal colon-separated (2001:db8::1), built-in IPSec, auto-configuration (SLAAC), no NAT needed. Dual-stack runs both simultaneously during transition.

Q52 What is a loopback address?

Ans: 127.0.0.1 (IPv4) and ::1 (IPv6) are loopback addresses used to test the local TCP/IP stack without physical network transmission. 'ping 127.0.0.1' tests if TCP/IP is functional on the local machine. Failure indicates a TCP/IP stack issue.

Q53 What is SNMP?

Ans: SNMP (Simple Network Management Protocol, port 161/162) monitors and manages network devices (routers, switches, servers, printers). It uses MIBs (Management Information Bases) to define device data. SNMP v3 adds encryption and authentication. Tools: Nagios, SolarWinds, PRTG.

Q54 What is the difference between half-duplex and full-duplex?

Ans: Half-duplex: data flows in one direction at a time — like a walkie-talkie. Legacy hubs use half-duplex. Full-duplex: data flows in both directions simultaneously. Modern switches support full-duplex, effectively doubling bandwidth. Duplex mismatch between switch port and NIC causes slow performance.

Q55 What is a MAC address table (CAM table)?

Ans: A switch maintains a CAM (Content Addressable Memory) table mapping MAC addresses to physical ports. When a frame arrives, the switch records source MAC + port. Frames are forwarded only to the correct port. Unknown destination MACs are flooded to all ports until learned.

Q56 What is spanning tree protocol (STP)?

Ans: STP (IEEE 802.1D) prevents Layer 2 broadcast storms by detecting and blocking redundant paths between switches. It elects a root bridge and puts redundant ports in blocking state. RSTP (Rapid STP) is the modern faster-converging version. Without STP, loops cause network failure.

Q57 What is a network topology?

Ans: Network topology describes device arrangement. Physical topologies: Bus (single cable, obsolete), Ring (tokens passed, obsolete), Star (all connect to central switch — most common), Mesh (all connected to all — high redundancy), Hybrid. Logical topology describes data flow path.

Q58 What is network documentation and why is it important?

Ans: Network documentation includes: network diagrams (physical and logical), IP addressing scheme, VLAN assignments, cable labels, rack diagrams, equipment inventory, and configuration backups. Essential for troubleshooting, onboarding new staff, disaster recovery, and compliance audits.

Q59 What is a cable tester and its uses?

Ans: A cable tester verifies Ethernet/coaxial cable integrity: checks for continuity on all 8 wires, detects opens (broken wires), shorts (two wires touching), crossed pairs, and split pairs. Advanced testers (TDR) measure cable length and locate fault distance. Essential for physical layer troubleshooting.

Q60 What is a protocol analyzer (packet sniffer)?

Ans: A protocol analyzer captures and analyzes network packets in real time. Wireshark is the most popular free tool. Used for: troubleshooting connectivity issues, analyzing performance, identifying malicious traffic, and verifying protocols. Requires NIC in promiscuous mode to capture all traffic.

SECTION 03

Hardware & Printers

PC Components, Storage, Power, Printers, Peripherals

Q61 – Q90

Q61 What is the difference between DDR4 and DDR5 RAM?

Ans: DDR4: 2133-3200 MHz base, 1.2V, dual-channel with 64-bit bus, JEDEC standard. DDR5: 4800-8400+ MHz, 1.1V, improved channel architecture (two 32-bit subchannels per DIMM), on-die ECC, higher density per module. DDR5 is not backward compatible with DDR4 slots.

Q62 What is ECC RAM and when is it used?

Ans: ECC (Error Correcting Code) RAM detects and corrects single-bit memory errors on the fly, preventing data corruption. Used in servers, workstations, and critical systems where data integrity is paramount. Requires ECC-capable motherboard and CPU (AMD EPYC/Ryzen Pro, Intel Xeon). Slightly slower than non-ECC.

Q63 What are the main motherboard form factors?

Ans: ATX (305x244mm, standard desktop), Micro-ATX (244x244mm, smaller with fewer slots), Mini-ITX (170x170mm, compact builds), E-ATX (Extended ATX, server/HEDT). Form factor determines: case compatibility, number of expansion slots, RAM slots, and I/O options. Always match motherboard to case.

Q64 What is PCIe and what are its lane configurations?

Ans: PCIe (Peripheral Component Interconnect Express) is the standard expansion interface for GPU, NVMe SSD, sound cards, and network cards. Lane configurations: x1 (1 lane, ~1 GB/s), x4, x8, x16 (16 lanes, ~32 GB/s PCIe 4.0). Generations: PCIe 3.0, 4.0 (2x faster), 5.0 (4x faster than 3.0).

Q65 What is the purpose of a CMOS battery and what happens when it fails?

Ans: The CMOS battery (CR2032 lithium coin cell) powers the CMOS chip maintaining BIOS settings, hardware configuration, and real-time clock (RTC) when PC is off. Failure symptoms: system loses date/time on every shutdown, BIOS resets to defaults, possible CMOS checksum error on boot.

Q66 How do you safely handle computer components to avoid ESD damage?

Ans: ESD (Electrostatic Discharge) can permanently damage chips even from charges invisible to humans. Prevention: use anti-static wrist strap grounded to chassis, work on anti-static mat, handle PCBs by edges, store components in anti-static bags, never work on carpet, discharge yourself by touching metal case before handling.

TIP ESD safety is heavily tested on CompTIA A+ – always wear antistatic wrist strap!

Q67 What are the different CPU socket types used in modern systems?

Ans: Intel: LGA1200 (10th/11th gen), LGA1700 (12th/13th/14th gen), LGA4677 (Xeon Scalable servers). AMD: AM4 (Ryzen 1000-5000 PGA socket), AM5 (Ryzen 7000+ LGA socket), SP3/SP5 (EPYC server). Important: socket type determines CPU compatibility – always verify before purchase.

Q68 What is the difference between a mechanical HDD and SSD?

Ans: HDD: magnetic spinning platters, actuator arm read/write head, 5400-7200 RPM, ~100-150 MB/s, affordable per GB, sensitive to vibration/drops, noisy. SSD: NAND flash memory, no moving parts, 500-7000+ MB/s (NVMe), silent, shock-resistant, faster boot/load times, higher cost per GB.

Q69 What is RAID and which levels should an A+ technician know?

Ans: RAID combines multiple drives. RAID 0: striping – improved speed, NO redundancy, failure loses all data. RAID 1: mirroring – full redundancy, 50% storage efficiency. RAID 5: striping with distributed parity – minimum 3 drives, tolerates 1 failure. RAID 10: mirror of stripes – fast + redundant, minimum 4 drives.

TIP Know RAID 0, 1, 5, 10 for the A+ exam. RAID 0 = no fault tolerance!

Q70 What is an NVMe SSD vs SATA SSD?

Ans: SATA SSD: connects via SATA III (max 600 MB/s), uses AHCI protocol designed for HDDs. NVMe (Non-Volatile Memory Express) SSD: connects via PCIe (M.2 or U.2), uses NVMe protocol optimized for flash, ~3500-7000 MB/s sequential read. NVMe is 6-12x faster than SATA SSD.

Q71 What PSU connector types are used in modern PCs?

Ans: 24-pin ATX (motherboard main power), 4/8-pin EPS/ATX12V (CPU power), 6/8-pin PCIe (GPU power), SATA power (15-pin, for SATA drives), Molex (4-pin legacy, older drives/fans), PCIe 12VHPWR (16-pin, RTX 4000 series GPUs). Always ensure PSU wattage is adequate for all components.

Q72 What is the 80 Plus PSU efficiency rating?

Ans: 80 Plus ratings indicate PSU efficiency (power out vs power consumed from wall). Ratings: 80 Plus (80%), Bronze (82-85%), Silver (85-88%), Gold (87-90%), Platinum (90-92%), Titanium (92-96% at 50% load). Higher efficiency = less heat, lower electricity bill, more reliable power delivery.

Q73 What is the difference between an inkjet and laser printer?

Ans: Inkjet: uses liquid ink cartridges, sprays droplets through microscopic nozzles, excellent photo quality, lower upfront cost, higher per-page cost, slower, can smear when wet. Laser: uses toner powder fused by heat drum, faster, sharp text, lower per-page cost for high volume, better for office use.

Q74 What are the steps of the laser printer imaging process?

Ans: 7 steps: 1) Processing (RIP converts data to bitmap), 2) Charging (corona wire/roller charges drum to -600V), 3) Exposing (laser writes image, reducing charge to -100V), 4) Developing (toner adheres to low-charge areas), 5) Transferring (toner moves to paper), 6) Fusing (heat+pressure bonds toner), 7) Cleaning (residual toner removed from drum).

TIP *The 7-step laser printing process is a guaranteed exam question on Core 1!*

Q75 What is a printer toner cartridge vs drum unit?

Ans: Toner cartridge contains the toner powder — replaced when prints fade or low-toner warning. Drum unit (photoconductor drum) is the light-sensitive cylinder that forms the latent image — lasts 20,000-50,000 pages. Some cartridges include the drum; others are separate. Drum issues cause streaks or ghosting.

Q76 What is a thermal printer and its applications?

Ans: Thermal printers use heat-sensitive paper or ribbon — no ink or toner. Types: Direct thermal (applies heat directly to paper — fades over time, used for receipts, labels). Thermal transfer (heat melts ribbon wax/resin onto paper/label — more durable). Used in retail POS, shipping labels, barcode printing.

Q77 What is a 3D printer?

Ans: 3D printers create three-dimensional objects from digital models using additive manufacturing — building layer by layer. Types: FDM (Fused Deposition Modeling, melts plastic filament), SLA (Stereolithography, UV-cured resin), SLS (Selective Laser Sintering, powder). Used in prototyping, manufacturing, and medical applications.

Q78 How do you troubleshoot a printer that produces faded prints?

Ans: Faded prints: low toner/ink (replace cartridge), toner saver mode enabled (disable in settings), wrong paper type (use compatible paper), dirty fuser (clean or replace), drum nearing end of life (replace drum unit). Test with printer self-test page to isolate whether it is a driver/computer issue.

Q79 What causes vertical streaks on laser printer output?

Ans: Vertical streaks (lines running the length of the page) indicate: scratched or worn drum — replace drum unit; contaminated or damaged fuser roller — clean or replace fuser; debris in toner cartridge — replace cartridge; dirty corona wire — carefully clean with provided cleaning tool.

Q80 What is a print server?

Ans: A print server manages print requests from multiple client computers and queues them to one or more printers. Can be: dedicated hardware device (network print server), software running on a server/PC (Windows Print Server), or built into a printer (via Ethernet or Wi-Fi). Centralizes printer management.

Q81 What is the purpose of a cooling system in a PC?

Ans: Cooling removes heat generated by CPU, GPU, RAM, and VRMs (voltage regulators). Methods: air cooling (heatsink + fan, case fans), AIO liquid cooling (all-in-one closed loop), custom water cooling. Thermal paste fills gaps between CPU and heatsink. Poor cooling causes throttling, instability, and reduced component lifespan.

Q82 What is an M.2 slot and what types of drives fit in it?

Ans: M.2 is a compact PCB slot on the motherboard for expansion cards. Supports two types: M.2 SATA (keyed B+M, max 600 MB/s) and M.2 NVMe PCIe (keyed M, up to 7000+ MB/s). A slot may support one or both types – check motherboard specs. Common lengths: 2242, 2260, 2280 (most common).

Q83 What is a sound card and when would you install a dedicated one?

Ans: A sound card (audio interface) processes audio I/O. Modern motherboards include onboard audio (Realtek codec) – adequate for most users. Dedicated sound cards are installed for: professional audio recording/production (low latency, high SNR), audiophile listening (higher DAC quality), or when onboard audio is faulty.

Q84 What is the purpose of a capture card?

Ans: A capture card records or streams video input from game consoles, cameras, or other devices to a computer. Available as PCIe internal cards or USB external devices. Essential for content creators streaming console gameplay. Examples: Elgato HD60, AVerMedia, Magewell.

Q85 What is the difference between USB 2.0, 3.0, and 3.1?

Ans: USB 2.0 (Hi-Speed): 480 Mbps, black port. USB 3.0/3.1 Gen 1 (SuperSpeed): 5 Gbps, blue port. USB 3.1 Gen 2 (SuperSpeed+): 10 Gbps, teal port. USB 3.2 Gen 2x2: 20 Gbps. USB4: 40 Gbps, USB-C only. Backward compatible – USB 3.x device works in USB 2.0 port at USB 2.0 speed.

Q86 What is a KVM switch?

Ans: KVM (Keyboard, Video, Mouse) switch allows one keyboard, monitor, and mouse to control multiple computers. Switch between systems via a button press or keyboard hotkey. Used in server rooms and by users managing multiple machines. HDMI/DisplayPort KVM switches support 4K output.

Q87 What is the purpose of a UPS and how do you size one?

Ans: UPS (Uninterruptible Power Supply) provides battery backup during power outages, protecting against data loss and hardware damage. Sizing: add wattage of all connected devices; UPS VA rating should be 25% more than total wattage. Runtime depends on battery capacity and load.

Q88 What is a surge protector and how does it differ from a UPS?

Ans: A surge protector absorbs excess voltage spikes to protect equipment — it has no battery backup. A UPS includes battery backup plus surge protection. All UPS units include surge protection; most surge protectors do not include battery backup. Clamping voltage determines surge protector quality.

Q89 What is a smart card reader used for in enterprise?

Ans: Smart card readers read embedded microprocessors in smart cards for authentication. Used for: physical access (building entry), logical access (computer login — CAC/PIV cards), digital signatures, secure email. Smart card + PIN = two-factor authentication. Required in government and military environments.

Q90 What is biometric hardware?

Ans: Biometric hardware captures and verifies unique physical characteristics for authentication. Types: fingerprint reader (capacitive), facial recognition camera (IR + visible light), iris scanner, vein pattern reader, voice recognition microphone. More convenient than passwords; cannot be forgotten but can be spoofed if poorly implemented.

SECTION 04

Virtualization & Cloud Computing

Hypervisors, Cloud Models, Containers, Remote Access

Q91 – Q120

Q91 What is virtualization and what problem does it solve?

Ans: Virtualization uses software (hypervisor) to create multiple virtual machines (VMs) on one physical server, each with isolated OS and resources. Solves: server underutilization (typical physical server uses 5-15% resources), reduces hardware costs, improves disaster recovery, enables testing without additional hardware.

Q92 What is the difference between Type 1 and Type 2 hypervisors?

Ans: Type 1 (bare-metal): runs directly on hardware without a host OS — maximum performance. Examples: VMware ESXi, Microsoft Hyper-V (server), Citrix XenServer. Type 2 (hosted): runs as an application on top of a host OS — easier to set up but less efficient. Examples: VMware Workstation, Oracle VirtualBox, Parallels.

TIP *Type 1 = server virtualization. Type 2 = desktop/lab virtualization. Know both for A+ exam.*

Q93 What is a virtual machine (VM)?

Ans: A VM is a software emulation of a physical computer running its own OS and applications, isolated from other VMs on the same physical host. Each VM has virtualized CPU, RAM, disk (VMDK/VHD files), and network adapter. VMs can be snapshotted, cloned, and migrated between hosts.

Q94 What is a snapshot in virtualization?

Ans: A snapshot captures the exact state of a VM (disk, memory, settings) at a point in time. Used for: safe testing of updates/software, quick rollback if changes cause issues, creating VM templates. Note: long-term snapshots consume storage and degrade VM performance.

Q95 What is containerization and how does Docker work?

Ans: Containers package application code + dependencies into isolated units sharing the host OS kernel — lighter than VMs (no guest OS). Docker is the leading container platform: Dockerfile defines the image, docker build creates it, docker run starts a container. Images stored in Docker Hub registry.

Q96 What is the difference between a VM and a container?

Ans: VM: full OS, hardware-level isolation, heavy (GB-sized), slow to start (minutes), higher security isolation. Container: shares host OS kernel, process-level isolation, lightweight (MB-sized), starts in seconds, slightly less isolated. VMs suit multi-OS workloads; containers suit microservices and DevOps pipelines.

Q97 What are the three cloud service models?

Ans: IaaS (Infrastructure as a Service): you manage OS and apps, provider manages hardware — AWS EC2, Azure VMs. PaaS (Platform as a Service): provider manages OS and runtime, you manage apps — Heroku, Google App Engine. SaaS (Software as a Service): fully managed application — Microsoft 365, Salesforce, Zoom.

Q98 What are the four cloud deployment models?

Ans: Public cloud: shared infrastructure, provider-managed, pay-as-you-go — AWS, Azure, GCP. Private cloud: dedicated infrastructure for one organization, on-premises or hosted. Community cloud: shared by organizations with common concerns (government). Hybrid cloud: combination of public + private with data portability between them.

Q99 What is cloud storage and its benefits?

Ans: Cloud storage stores data on remote servers accessed via internet. Benefits: anywhere access, automatic redundancy (data replicated across multiple locations), elastic scaling, collaboration, and cost efficiency (pay per GB used). Examples: OneDrive, Google Drive, Dropbox, AWS S3, Azure Blob Storage.

Q100 What is a VDI (Virtual Desktop Infrastructure)?

Ans: VDI hosts desktop operating systems on centralized servers in the data center; users access them remotely via thin clients or regular PCs. Benefits: centralized management, data stays in data center (secure), easy deployment of OS updates. Used in call centers, healthcare, education. Examples: VMware Horizon, Citrix Virtual Apps.

Q101 What is RDP (Remote Desktop Protocol)?

Ans: RDP (port 3389) is Microsoft's protocol for connecting to Windows desktops/servers remotely, transmitting screen, keyboard, and mouse data. Provides full interactive session. Security: use VPN + NLA (Network Level Authentication), change default port, restrict access by IP. Client: mstsc.exe on Windows.

TIP RDP port 3389 appears frequently on A+ Core 2 security exam questions.

Q102 What is SSH and when is it used?

Ans: SSH (Secure Shell, port 22) provides encrypted command-line access to remote systems, primarily Linux/Unix servers and network equipment. Replaces insecure Telnet (port 23). SSH also supports SFTP (secure file transfer) and port forwarding/tunneling. Key-based authentication is more secure than passwords.

Q103 What is a thin client?

Ans: A thin client is a lightweight endpoint device with minimal local processing that connects to a remote server (VDI or terminal services) for computing. Low cost, low power, easy management, long lifespan. No local data storage – all data lives on server. Ideal for VDI deployments and high-security environments.

Q104 What is Microsoft Azure and AWS from a technician's perspective?

Ans: Azure (Microsoft) and AWS (Amazon) are the two leading public cloud platforms. From an A+ perspective: Azure AD (now Entra ID) for cloud identity management, Azure VMs, OneDrive/SharePoint for storage. AWS EC2 for compute, S3 for storage. CompTIA A+ tests cloud concepts, not deep cloud administration.

Q105 What is Infrastructure as Code (IaC)?

Ans: IaC manages and provisions cloud infrastructure using machine-readable configuration files instead of manual processes. Tools: Terraform, AWS CloudFormation, Azure Resource Manager. Benefits: consistency, version control (Git), automated deployment, and documented infrastructure. Reduces human error.

Q106 What is a content delivery network (CDN)?

Ans: A CDN distributes web content from servers geographically close to users, reducing latency and improving load times. Caches static assets (images, CSS, JS, video). Also provides DDoS protection and SSL termination. Examples: Cloudflare, AWS CloudFront, Akamai. Transparent to end users.

Q107 What is cloud bursting?

Ans: Cloud bursting is a configuration where an on-premises (private cloud) infrastructure automatically scales into a public cloud when local resources reach capacity. The overflow workload 'bursts' to the cloud temporarily, then returns on-premises when demand decreases. Cost-effective for variable workloads.

Q108 What is multi-tenancy in cloud computing?

Ans: Multi-tenancy means multiple customers (tenants) share the same physical hardware and software infrastructure but are logically isolated. Each tenant's data is separated and secure. The cloud provider ensures one tenant cannot access another's data. Lower cost than dedicated single-tenant infrastructure.

Q109 What is Software Defined Networking (SDN)?

Ans: SDN separates the network control plane (decision-making) from the data plane (traffic forwarding), centralizing control in software controllers. Benefits: programmable networks, dynamic configuration, automation, and visibility. Used in data centers and cloud providers for flexible, automated network management.

Q110 What is a cloud SLA and what metrics matter?

Ans: Cloud SLAs define: uptime guarantee (e.g., 99.9% = max 8.77 hours downtime/year, 99.99% = 52 minutes), RTO (Recovery Time Objective), RPO (Recovery Point Objective), support response times, and credit compensation for violations. Always review SLA before selecting a cloud provider.

Q111 What is disaster recovery as a service (DRaaS)?

Ans: DRaaS provides cloud-based replication and recovery of on-premises systems. If primary site fails, workloads fail over to cloud VMs within minutes. Cost-effective alternative to traditional DR (second physical data center). Providers: AWS Elastic Disaster Recovery, Zerto, Veeam Cloud Connect.

Q112 What is a virtual NIC (vNIC)?

Ans: A virtual NIC is a software emulation of a physical network interface card assigned to a virtual machine. The hypervisor connects vNICs to virtual switches (vSwitches) which then connect to physical NICs. VMs can have multiple vNICs on different virtual networks for traffic isolation.

Q113 What is live migration of VMs?

Ans: Live migration (vMotion in VMware, Live Migration in Hyper-V) moves a running VM from one physical host to another with zero or minimal downtime. The VM's memory state is transferred while it continues running. Used for host maintenance, load balancing, and hardware upgrades without service interruption.

Q114 What is a virtual switch (vSwitch)?

Ans: A virtual switch is software simulating a physical Layer 2 network switch within a hypervisor. It connects VMs on the same host to each other and to external physical networks via physical NICs. Supports VLANs, port groups, security policies, and traffic shaping.

Q115 What is GPU passthrough in virtualization?

Ans: GPU passthrough (PCIe passthrough) dedicates a physical GPU directly to a single VM, bypassing the hypervisor's virtual GPU emulation. Provides near-native GPU performance for gaming VMs, ML training, or GPU-accelerated workloads. Requires IOMMU support (Intel VT-d / AMD-Vi) in CPU and motherboard.

Q116 What is nested virtualization?

Ans: Nested virtualization allows running a hypervisor inside a VM — enabling a VM to host its own VMs. Useful for testing hypervisors, cloud labs, and training environments. Must be explicitly enabled on the host hypervisor. Performance overhead is significant — not for production use.

Q117 What is auto-scaling in cloud computing?

Ans: Auto-scaling automatically adjusts the number of computing resources (VMs, containers) based on demand metrics (CPU usage, request rate). Scale out: add instances when load increases. Scale in: remove instances when load decreases. Reduces cost (pay for what you use) and ensures performance during traffic spikes.

Q118 What is a virtual private cloud (VPC)?

Ans: A VPC is a logically isolated section of a public cloud where you launch resources in a virtual network you define. You control: IP address ranges, subnets, routing tables, security groups (firewall rules), and gateways. Each cloud provider has a VPC equivalent: AWS VPC, Azure Virtual Network (VNet), GCP VPC.

Q119 What is edge computing in relation to cloud?

Ans: Edge computing processes data near its source (IoT sensors, factory floors, retail stores) rather than sending everything to a central cloud data center. Reduces latency for real-time decisions, conserves bandwidth, and improves reliability when cloud connectivity is intermittent. Complements — not replaces — cloud.

Q120 What security considerations apply to cloud environments?

Ans: Key cloud security concerns: shared responsibility model (provider secures infrastructure; customer secures data, access, and applications), data sovereignty (where data is stored legally), access management (IAM — Identity and Access Management), encryption at rest and in transit, audit logging, and misconfigured storage buckets.

SECTION 05

Hardware Troubleshooting

Diagnostics, POST, BSOD, Storage, Display, Power Issues

Q121 – Q150

Q121 What is the first step in the CompTIA troubleshooting methodology?

Ans: CompTIA's 6-step methodology: 1) Identify the problem (symptoms, recent changes, question the user), 2) Establish a theory of probable cause, 3) Test the theory to determine the cause, 4) Establish a plan of action and implement the solution, 5) Verify full system functionality, 6) Document findings, actions, and outcomes.

TIP *This exact 6-step process is tested directly on Core 2 – memorize it!*

Q122 What is POST and what do beep codes indicate?

Ans: POST (Power-On Self-Test) is BIOS/UEFI's hardware diagnostic run at every startup. Beep codes vary by BIOS manufacturer (AMI, Award, Phoenix). Common: 1 short beep = POST OK; continuous beeps = power issue; 3 beeps = RAM error; no beep + no display = dead CPU/motherboard. LED diagnostic codes on modern boards supplement beeps.

Q123 What is a Blue Screen of Death (BSOD) and how do you diagnose it?

Ans: BSOD (Windows stop error) shows a stop code when the kernel encounters a fatal error it cannot recover from. Diagnosis: note the stop code (e.g., IRQL_NOT_LESS_OR_EQUAL = driver issue, MEMORY_MANAGEMENT = RAM issue), check Event Viewer (System log), analyze minidump files with WinDbg or WhoCrashed.

Q124 What does a PC that powers on but has no display indicate?

Ans: Possible causes: display cable disconnected/faulty, monitor not on or wrong input selected, GPU seated improperly (reseat GPU), RAM not seated (reseat all sticks), CPU not seated/bent pins, BIOS needs update for new CPU, onboard graphics disabled in BIOS. Test with a known-good monitor and cable.

Q125 What causes a PC to randomly restart?

Ans: Random restarts: overheating (CPU/GPU fan failure, dust blockage — clean and apply new thermal paste), failing PSU (insufficient/unstable power — test with PSU tester), RAM failure (run MemTest86+), driver issues, Windows Update failure, malware. Check Event Viewer for critical errors before each restart.

Q126 What does it mean when a hard drive makes clicking sounds?

Ans: Clicking sounds from an HDD (click of death) indicate read/write head failure, damaged platters, or head actuator arm fault. The drive is failing immediately. Actions: stop using drive to prevent further damage, attempt data recovery immediately (professional data recovery if important), replace drive, restore from backup.

TIP *A clicking HDD = imminent death. Back up data IMMEDIATELY!*

Q127 What is S.M.A.R.T. and how does it help in disk troubleshooting?

Ans: S.M.A.R.T. (Self-Monitoring, Analysis and Reporting Technology) is a monitoring system in HDDs and SSDs reporting health attributes: reallocated sectors, spin-up time, read errors, temperature. Tools: CrystalDiskInfo, Speccy, manufacturer diagnostics. Pre-failure warning allows proactive replacement before data loss.

Q128 How do you troubleshoot a slow computer?

Ans: Systematic approach: check Task Manager (CPU/RAM/disk/network usage), look for high disk usage (disable Windows Search indexing if 100% disk), scan for malware, check startup programs (disable unneeded), check available disk space (under 15% free degrades performance), run chkdsk, consider hardware upgrade (RAM or SSD).

Q129 What is chkdsk and how is it run?

Ans: chkdsk scans and repairs file system errors and bad sectors. Command: chkdsk C: /f /r /x — /f fixes errors, /r locates bad sectors and recovers readable data, /x forces dismount. Requires restart for system drive. Run from elevated Command Prompt. Results in Event Viewer under Winit log.

Q130 What is the SFC /scannow command?

Ans: SFC (System File Checker) scans Windows protected system files for corruption and replaces them from the Windows component store cache. Run as administrator: 'sfc /scannow'. If SFC reports errors it cannot fix, run DISM first: 'DISM /Online /Cleanup-Image /RestoreHealth' to repair the component store, then run SFC again.

Q131 What does a burning smell from a PC indicate?

Ans: Burning smell indicates: PSU failing (capacitors overheating — stop use immediately, fire risk), burned motherboard component, overheated CPU/GPU, short circuit. If burning smell is detected: immediately power off the system, unplug from wall, wait for cooling, inspect for visibly burned components, do not restart until fault is found.

Q132 What causes no sound on a computer and how do you troubleshoot?

Ans: Troubleshooting audio: check speaker/headphone connections, check volume level and mute status, verify correct playback device selected (right-click speaker icon), update/reinstall audio driver, check Device Manager for audio device errors, test with different speakers/headphones, verify audio service running (Windows Audio service).

Q133 What does 'Operating System not found' error mean?

Ans: This error means BIOS cannot find a bootable OS. Causes: wrong boot order in BIOS (check boot sequence — storage device with OS must be first), corrupted MBR/GPT boot record, failed storage device, OS files deleted. Fixes: fix boot order, run Startup Repair from Windows installation media, rebuild BCD.

Q134 How do you troubleshoot overheating?

Ans: Check CPU/GPU temperatures with HWMonitor or Core Temp. Overheating: clean dust from heatsinks and fans with compressed air, verify all fans are spinning (check BIOS fan monitor), replace thermal paste (often dried out on laptops/older PCs), ensure adequate case airflow (intake front/bottom, exhaust rear/top), verify thermal throttling isn't occurring.

Q135 What is a POST card and how is it used?

Ans: A POST diagnostic card plugs into a PCIe or USB slot and displays 2-digit hexadecimal POST codes as the system initializes. When system fails to boot with no beeps/display, POST card identifies exactly where initialization fails. Cross-reference code with motherboard manual for component-level diagnosis.

Q136 What common display issues does a technician encounter?

Ans: Dim/no backlight (inverter failure on CCFL; LED driver failure), flickering (loose cable, inverter, or GPU issue), dead pixels (physical LCD damage), color distortion (GPU failing, cable damage), incorrect resolution (driver issue, wrong monitor profile), burn-in (OLED/plasma; use screensaver, reduce brightness).

Q137 What is memory testing and how do you perform it?

Ans: Memory testing identifies faulty RAM modules. Tools: Windows Memory Diagnostic (mdsched.exe – runs on reboot), MemTest86 (bootable USB, most thorough – run for 2+ passes). Isolate failed modules by testing one at a time. Symptoms of bad RAM: BSODs (MEMORY_MANAGEMENT), random crashes, application errors.

Q138 How do you troubleshoot USB devices not recognized?

Ans: Try: different USB port (rule out dead port), different computer (rule out device failure), update USB host controller driver, uninstall/reinstall device in Device Manager, disable USB selective suspend in Power Options, check USB power delivery (powered hub for power-hungry devices), check for conflicting software.

Q139 What is the beep code for RAM failure on AMI BIOS?

Ans: AMI BIOS RAM error beep codes: 3 short beeps = memory error (3-4-1 or 3-3-4 pattern), all-short continuous beeps indicate critical memory fault. Other manufacturers differ – check motherboard documentation. Modern boards often show Q-LED indicators (CPU/DRAM/VGA/BOOT) for post failure location.

Q140 What is thermal paste and when should it be replaced?

Ans: Thermal paste fills microscopic gaps between CPU and heatsink, dramatically improving heat transfer. Original paste dries out over 3-5 years, increasing temperatures. Replace when: CPU temperatures elevated, reseating cooler, upgrading CPU or cooler. Clean old paste with isopropyl alcohol 90%+, apply fresh paste (pea-sized for most CPUs).

Q141 What does 'No boot device found' error mean and how do you fix it?

Ans: System cannot find bootable storage. Fixes: enter BIOS and verify storage device appears in detected devices, correct boot order (NVMe SSD or SATA drive must be first boot device), check storage cable connections (SATA data + power), run startup repair from Windows media, check if drive is visible in Disk Management.

Q142 What is disk encryption and how does it affect troubleshooting?

Ans: BitLocker (Windows) or FileVault (macOS) encrypts entire drive, requiring a key to access data. If encryption key is lost, data is unrecoverable. Troubleshooting encrypted drives: always save BitLocker recovery key to Microsoft account or AD before encrypting, enter recovery key when prompted during hardware changes.

Q143 What are signs of PSU failure?

Ans: PSU failure signs: system won't power on (no POST), random shutdowns/restarts under load, instability with multiple components, burning smell from PSU, coil whine, incorrect voltages measured on 24-pin connector. Test PSU with PSU tester or multimeter. Replace rather than repair – PSU failure can damage other components.

Q144 What is a paper jam in a printer and how do you clear it?

Ans: Paper jams occur from: overfilling paper tray, wrong paper size/type, worn pickup rollers, debris in paper path, damp/wavy paper. Clearing: follow printer manual, open all access doors, remove paper gently without tearing (torn pieces cause future jams), clean rollers with slightly damp cloth, fan paper before loading.

Q145 What are symptoms of a failing GPU?

Ans: GPU failure symptoms: display artifacts (colored dots, lines, polygons on screen), screen flickering, driver crashes with black screen, display going blank during gaming, high GPU temperatures with thermal throttling, computer crashing during graphically intensive tasks. Test by swapping GPU, running GPU stress test (FurMark).

Q146 How do you diagnose a laptop that won't charge?

Ans: Troubleshooting: check AC adapter LED (if present), try different outlet, inspect charging port for debris/damage, try compatible AC adapter, verify battery condition (BIOS battery report), check power management settings, update/reset BIOS (some have charging circuit firmware). If DC jack loose — repair required.

Q147 What is a loopback plug test?

Ans: A loopback plug (loopback adapter) connects TX (transmit) to RX (receive) on a port, sending its own signal back to itself. Used to test NIC port functionality without needing an external connection. If loopback test passes but connectivity fails, the issue is external (cable, switch port, network). Available for RJ45, serial, USB.

Q148 What is a POST code and how does it differ from a beep code?

Ans: POST codes are hexadecimal values displayed on a 2-digit LED display (onboard or POST card) indicating initialization stage. Beep codes are audible patterns from the system speaker. Both diagnose boot failures, but POST codes are more precise (256 possible values vs a few beep patterns). Modern motherboards increasingly use LED codes.

Q149 What tools are essential in a hardware technician's toolkit?

Ans: Essential tools: Phillips/flathead screwdrivers (magnetic), anti-static wrist strap, ESD mat, cable ties, cable tester, multimeter, flashlight/headlamp, tweezers/parts grabber, USB bootable diagnostic drives (MemTest86, WinPE), thermal paste, isopropyl alcohol wipes, compressed air, and a POST/diagnostic card.

Q150 What is the process for replacing a laptop screen?

Ans: Process: power off and remove battery, consult service manual for display disassembly, remove display bezel (pry clips carefully), disconnect LVDS/eDP display cable and (if present) Wi-Fi antenna cables, unscrew screen from hinges, install new screen reversing steps, reconnect cables, test before securing bezel.

SECTION 06

Operating Systems

Windows, Linux, macOS, Command Line, File Systems, OS Management

Q151 – Q180

Q151 What editions of Windows 10/11 does A+ cover?

Ans: Windows Home: basic features, no domain join, no Group Policy, no BitLocker, no Remote Desktop host. Windows Pro: adds domain join, Group Policy, BitLocker, Remote Desktop host, Hyper-V, assigned access. Windows Pro for Workstations: adds ReFS, persistent memory, more RAM. Windows Enterprise: adds all Pro features plus advanced security (WDAG, AppLocker).

Q152 What is the Windows registry and when should you edit it?

Ans: The Registry is a hierarchical database storing OS and application settings. Key hives: HKLM (Local Machine – hardware/OS settings), HKCU (Current User – per-user settings), HKCR (Classes Root – file associations), HKU (Users), HKCC (Hardware profile). Edit only when necessary (regedit.exe); always backup first (File > Export).

TIP *Know the 5 registry hives for the A+ exam – especially HKLM and HKCU!*

Q153 What is the Windows Task Manager and what can it do?

Ans: Task Manager (Ctrl+Shift+Esc): Processes tab (running apps + background processes, CPU/memory/disk/network per process, right-click to kill), Performance tab (CPU, memory, disk, GPU graphs), Startup tab (disable startup programs – CRITICAL for slow boot), Services tab, Details tab (PID, user, priority).

Q154 What are the key Windows administrative tools?

Ans: Device Manager (devmgmt.msc), Disk Management (diskmgmt.msc), Computer Management (compmgmt.msc), Services (services.msc), Event Viewer (eventvwr.msc), Local Group Policy Editor (gpedit.msc – Pro/Enterprise only), Registry Editor (regedit.exe), Task Scheduler (taskschd.msc), Performance Monitor (perfmon.msc).

Q155 What is NTFS and what features does it have over FAT32?

Ans: NTFS features: file sizes up to 16 TB, volume size up to 8 PB, file/folder permissions (ACLs), encryption (EFS), compression, journaling (prevents corruption), alternate data streams, disk quotas, shadow copies (VSS). FAT32 limitations: 4 GB max file size, 8 TB max volume, no permissions/journaling.

Q156 What is the Windows boot process?

Ans: UEFI/BIOS □ POST □ Boot device selected □ GPT/MBR read □ Windows Boot Manager (bootmgr) □ Boot Configuration Data (BCD) loaded □ Windows OS Loader (winload.efi) □ NTOSKRNL.EXE (kernel) loads □ SMSS.EXE (session manager) □ WININIT.EXE □ SVCHOST + services start □ Login screen (winlogon.exe).

Q157 What is the difference between FAT32, exFAT, and NTFS?

Ans: FAT32: max 4 GB file, 8 TB volume, no permissions, compatible with all devices — best for USB drives shared across OSes. exFAT: large file support, no permissions, designed for flash storage — best for large USB drives/SD cards. NTFS: advanced features, max file/volume size, permissions — required for Windows boot drives.

Q158 What is the Windows command prompt and key commands?

Ans: Key commands: ipconfig (network config), ping (test connectivity), tracert (trace route), netstat (network connections), nslookup (DNS query), chkdsk (disk check), sfc (system file check), DISM (component store repair), tasklist/taskkill (process management), net user/net group (user management), robocopy (robust file copy).

Q159 What is PowerShell and how does it differ from CMD?

Ans: PowerShell is an advanced scripting environment built on .NET, working with objects rather than text. More powerful than CMD for automation, system administration, and scripting. Can run CMD commands plus PowerShell cmdlets (verb-noun format, e.g., Get-Service, Set-ExecutionPolicy). PowerShell ISE / VS Code for scripting.

Q160 What is the Windows Safe Mode and its variants?

Ans: Safe Mode loads minimal drivers and services — useful for removing malware, fixing driver issues. Variants: Safe Mode (minimal), Safe Mode with Networking (adds networking — for downloading fixes), Safe Mode with Command Prompt (no GUI). Access: hold Shift + Restart □ Troubleshoot □ Advanced Options □ Startup Settings □ F4/F5/F6.

Q161 What is System Restore and how does it work?

Ans: System Restore creates restore points (snapshots of system files, registry, and installed programs) before major changes. If problems occur, roll back to a previous restore point. Does NOT affect personal files. Managed via System Properties □ System Protection. Disable on drives with limited space.

Q162 What is the Windows Event Viewer and its key logs?

Ans: Event Viewer logs: Application (app errors/crashes), Security (logon events, policy changes, auditing), System (OS and driver events), Setup, Forwarded Events. Levels: Critical, Error, Warning, Information, Verbose. Use to: diagnose BSODs (look for events before crash), audit logins, diagnose service failures.

Q163 What is UAC (User Account Control)?

Ans: UAC prompts for confirmation or admin credentials when actions require elevated privileges (even for admin accounts) — limiting malware damage. Levels: Always Notify (most secure), Notify only for app changes, Notify only for app changes without dimming desktop, Never Notify (insecure). Change via User Accounts control panel.

Q164 What is BitLocker and how does it work?

Ans: BitLocker is full-disk encryption built into Windows Pro/Enterprise/Education. It encrypts the entire drive with AES-128 or AES-256. Requires TPM 1.2+ for key storage. Recovery key must be saved (Microsoft account, USB, or printout). Transparent operation — no performance impact for SSDs. Protects data if drive is removed.

Q165 What is Windows Defender and its components?

Ans: Windows Defender / Microsoft Defender is the built-in security platform: Antivirus (real-time malware protection), Firewall (stateful network firewall), SmartScreen (URL/download filtering), Credential Guard (virtualization-based credential protection), Device Guard (application control), Attack Surface Reduction (ASR) rules.

Q166 What is the difference between 32-bit and 64-bit Windows?

Ans: 32-bit Windows: max 4 GB RAM, runs only 32-bit apps. 64-bit Windows: supports terabytes of RAM, runs both 32-bit (in WOW64 compatibility layer) and 64-bit apps. 64-bit apps run in C:\Program Files; 32-bit apps in C:\Program Files (x86). All modern hardware and Windows versions are 64-bit.

Q167 What are the Linux distributions a technician should know?

Ans: CompTIA A+ covers Linux basics. Key distros: Ubuntu (Debian-based, most popular desktop, good for beginners), Red Hat Enterprise Linux/CentOS/Rocky (enterprise, certification-focused), Kali Linux (security testing), Raspberry Pi OS (embedded). Know: package managers (apt for Debian, yum/dnf for Red Hat), /etc for config files.

Q168 What are essential Linux commands for A+ technicians?

Ans: ls (list files), cd (change directory), pwd (print working directory), mkdir/rmdir (create/remove directory), cp/mv/rm (copy/move/delete), cat/less/more (view files), grep (search text), chmod/chown (permissions/ownership), sudo (run as root), apt/yum (package management), ps/top/kill (process management), ifconfig/ip (network).

Q169 What is macOS and how does it differ from Windows for support?

Ans: macOS runs on Apple hardware (Mac). Key differences: HFS+/APFS file system (vs NTFS), uses /Users for home directories (vs C:\Users), Finder instead of File Explorer, Activity Monitor instead of Task Manager, Terminal (Unix-based) instead of CMD, System Preferences/Settings instead of Control Panel, no registry.

Q170 What is the Windows disk management tool and what can it do?

Ans: Disk Management (diskmgmt.msc): initialize new disks (MBR or GPT), create/delete/format partitions, assign drive letters, extend/shrink volumes, mark partition as active (boot), convert basic to dynamic disk, set up RAID (software). For command line: diskpart utility offers same functionality plus more advanced options.

Q171 What is a Windows user profile?

Ans: A user profile stores user-specific settings in C:\Users\[username]: Desktop, Documents, Downloads, AppData (application settings), NTUSER.DAT (user registry hive). Types: Local profile (on PC only), Roaming profile (copied to server, follows user to any domain PC — requires AD), Mandatory profile (read-only, changes not saved).

Q172 What is Group Policy and where is it used?

Ans: Group Policy (only on Pro/Enterprise) enforces settings for users/computers in a domain (via AD) or locally (Local Group Policy — gpedit.msc). Can configure: password policies, software restrictions, drive mappings, desktop lockdowns, security settings, Windows Update settings. Applied hierarchically: Local □ Site □ Domain □ OU.

Q173 What is Windows Update and how do you manage it?

Ans: Windows Update downloads and installs security patches, feature updates, driver updates, and cumulative updates. Manage via Settings □ Windows Update. Enterprise uses WSUS (Windows Server Update Services) or Microsoft Endpoint Configuration Manager for centralized control. WU delivery optimization uses P2P for bandwidth savings.

Q174 What is the difference between an upgrade and a clean install of Windows?

Ans: Upgrade: installs new Windows version over existing, preserving installed apps, user files, and most settings — in-place upgrade. Clean install: formats the drive and installs fresh Windows — removes all data, apps, and settings. Recommended for troubleshooting persistent issues and for major version changes.

Q175 What are Windows file/folder permissions (NTFS)?

Ans: NTFS permissions: Full Control (all operations), Modify (read/write/execute/delete), Read & Execute, List Folder Contents, Read, Write. Assigned to users or groups via security tab. Inheritance: child objects inherit parent permissions by default. Effective permissions = combination of all assigned permissions minus denies.

Q176 What is the Windows firewall?

Ans: Windows Defender Firewall is a host-based stateful firewall filtering inbound/outbound traffic based on rules. Profiles: Domain (corporate network), Private (home/trusted), Public (untrusted/hotspot – most restrictive). Configured via Control Panel, Settings, or Group Policy. wf.msc for advanced rules.

Q177 What is the Task Scheduler in Windows?

Ans: Task Scheduler (taskschd.msc) automates programs or scripts triggered by: specific times, system events (startup, login, idle, event log entries), or conditions (AC power only). Used for: automated backups, maintenance scripts, AV scans, system monitoring. Malware often creates tasks for persistence – check regularly.

Q178 What is a Windows image backup vs file backup?

Ans: System image backup creates a sector-by-sector copy of entire drive(s) – restores everything including OS and apps but requires same or larger target drive. File backup copies selected files/folders – more flexible restoration. Windows Backup + Restore (Windows 7) and File History handle both. For enterprise: Veeam, Backup Exec.

Q179 What is the PATH environment variable?

Ans: PATH is a system environment variable listing directories the OS searches when running commands without specifying a full file path. Example: python is in C:\Python310\, adding this to PATH allows running 'python' from any directory. View/edit: System Properties □ Advanced □ Environment Variables.

Q180 What is Virtual Memory and the pagefile?

Ans: Virtual memory extends RAM by using disk space as overflow memory. The pagefile (pagefile.sys on Windows) stores data from RAM that hasn't been used recently. Performance degrades when pagefile is heavily used (disk I/O is 100x slower than RAM). Default: system-managed size. Move pagefile to fastest drive.

SECTION 07

Security

Threats, Authentication, Encryption, Hardening, SOHO Security

Q181 – Q210

Q181 What are the CompTIA A+ security threat categories?

Ans: Social Engineering (phishing, vishing, tailgating, shoulder surfing), Malware (virus, worm, trojan, ransomware, rootkit, keylogger, spyware, adware, cryptominer), Physical threats (theft, dumpster diving, hardware keyloggers), Network attacks (DDoS, MITM, spoofing, brute force, SQL injection, XSS).

TIP *Security is ~24% of Core 2 exam – single largest domain!*

Q182 What is social engineering?

Ans: Social engineering manipulates people into revealing information or performing actions. Techniques: Phishing (fraudulent email), Vishing (voice/phone), Smishing (SMS), Spear phishing (targeted), Whaling (targeting executives), Tailgating/Piggybacking (physical entry), Pretexting (false identity/scenario), Dumpster diving, Shoulder surfing.

Q183 What is ransomware and how do you protect against it?

Ans: Ransomware encrypts victims' files and demands payment for decryption key. Protection: regular offline backups (3-2-1 rule), email filtering to block malicious attachments, patch management, user training, endpoint protection, restrict admin rights. Recovery: restore from clean backup – never pay ransom (no guarantee of key).

Q184 What is a rootkit?

Ans: A rootkit gains privileged (root/admin) access to a system while hiding its presence from users, antivirus, and the OS. It may hide files, processes, registry keys, or network connections. Extremely difficult to detect with conventional tools. Removal often requires bootable antivirus scanner or complete OS reinstallation.

Q185 What is a keylogger?

Ans: A keylogger records every keystroke, capturing passwords, credit card numbers, and messages. Types: software (runs in background, part of malware), hardware (physical device between keyboard and PC — check USB ports on public computers). Detected by antivirus; hardware keyloggers require physical inspection.

Q186 What is a man-in-the-middle (MITM) attack?

Ans: MITM intercepts communication between two parties without their knowledge, allowing the attacker to eavesdrop or modify data. Methods: ARP poisoning (LAN), rogue Wi-Fi hotspot, SSL stripping. Prevention: HTTPS everywhere, VPN, certificate pinning, network monitoring for ARP anomalies.

Q187 What is multifactor authentication (MFA) and why is it important?

Ans: MFA requires two or more authentication factors: something you know (password/PIN), something you have (hardware token, authenticator app, smart card), something you are (fingerprint, face ID), somewhere you are (location), something you do (behavioral). MFA makes stolen passwords insufficient for account compromise.

Q188 What is least privilege and why is it a security best practice?

Ans: Least privilege grants users only the minimum permissions necessary for their job function. Benefits: limits blast radius of compromised accounts, prevents accidental changes, reduces insider threat impact, limits malware propagation. Apply to: user accounts, service accounts, application permissions, database access.

Q189 What is the principle of need-to-know?

Ans: Need-to-know restricts information access to only those who require it for their specific duties, even if they have appropriate clearance level. Complements least privilege — clearance alone is insufficient; the person must also need the information. Standard in government/military and critical infrastructure.

Q190 What is encryption and its types?

Ans: Encryption converts plaintext to ciphertext. Symmetric: same key for encryption and decryption — fast, used for bulk data (AES-256). Asymmetric: public key encrypts, private key decrypts — slower, used for key exchange and digital signatures (RSA, ECC). Hashing: one-way transformation for integrity verification (SHA-256, MD5-deprecated).

Q191 What is SSL/TLS and how does HTTPS work?

Ans: TLS (Transport Layer Security, successor to SSL) encrypts data between browser and web server. HTTPS process: browser requests HTTPS, server presents digital certificate, browser verifies certificate with CA, TLS handshake establishes session keys (asymmetric), encrypted data transmission (symmetric AES). Certificate contains domain name + public key.

Q192 What is a digital certificate and Certificate Authority (CA)?

Ans: A digital certificate binds a public key to an entity's identity, signed by a CA. Certificate contains: subject (website/company), public key, validity period, CA signature. CA hierarchy: Root CA (self-signed, highly trusted) □ Intermediate CA □ End-entity certificate. Revoked certificates listed in CRL or checked via OCSP.

Q193 What is a VPN and its types?

Ans: VPN creates an encrypted tunnel over a public network. Types: Remote Access VPN (individual user connects to corporate network — IPsec or SSL/TLS, client software like Cisco AnyConnect, GlobalProtect), Site-to-Site VPN (connects two office networks permanently — no client software needed). Split tunneling sends only corporate traffic through VPN.

Q194 What is a firewall and what types exist?

Ans: Firewall types: Packet filtering (stateless, inspects headers only — basic), Stateful inspection (tracks connection state — better), Application-layer/Proxy (inspects content — DPI), NGFW (Next-Generation Firewall — includes IPS, application awareness, SSL inspection, user identity). Host-based (Windows Firewall) vs network-based (FortiGate, Palo Alto).

Q195 What is an IDS vs IPS?

Ans: IDS (Intrusion Detection System): monitors traffic passively and alerts on suspicious activity — does NOT block. IPS (Intrusion Prevention System): actively monitors and blocks malicious traffic inline. HIDS: host-based (monitors single system log/file changes). NIDS: network-based (monitors network traffic). Most NGFWs include IPS.

Q196 What is a SOHO network and how do you secure it?

Ans: SOHO (Small Office/Home Office) security best practices: change default router admin password, use WPA3 (or WPA2-AES minimum), disable WPS (PIN-based), enable guest network for visitors, update router firmware, disable remote management, use HTTPS for admin, enable firewall, change default SSID, use MAC filtering as additional layer.

Q197 What is port security on a network switch?

Ans: Port security limits the number of MAC addresses allowed on a switch port and can restrict access to specific MAC addresses. Violation modes: protect (drop unauthorized frames silently), restrict (drop + log), shutdown (disable port — requires admin to re-enable). Prevents MAC flooding and unauthorized devices.

Q198 What is data destruction and acceptable methods?

Ans: When decommissioning hardware: Overwriting (software tools like DBAN, DoD 5220.22-M standard — write 0s/1s multiple passes), Shredding (physical destruction — NIST 800-88 approved), Degaussing (magnetic field erases HDDs — ineffective on SSDs/optical), Incineration, Secure erase (manufacturer tools for SSDs). Document destruction for compliance.

TIP *Data destruction methods appear in A+ Core 2 security and operational procedures domains.*

Q199 What is an access control list (ACL)?

Ans: ACL is a set of rules controlling access to resources. Network ACLs (on routers/firewalls): permit or deny traffic based on source/destination IP, port, and protocol. NTFS ACLs: control file/folder permissions per user/group. Rules processed top-down — first match wins. Implicit deny at end of most ACLs.

Q200 What is DNS poisoning (cache poisoning)?

Ans: DNS cache poisoning injects malicious DNS records into a resolver's cache, redirecting users to fraudulent sites without their knowledge. Example: user requests bank.com but gets IP of phishing site. Prevention: DNSSEC (cryptographic DNS signing), use trusted DNS resolvers (1.1.1.1, 8.8.8.8), monitor DNS query logs.

Q201 What is a password attack and its types?

Ans: Password attack types: Brute force (try all combinations — slow but thorough), Dictionary attack (try common words/passwords), Rainbow table (precomputed hash lookup — defeated by salting), Credential stuffing (use leaked username/password pairs on other sites), Spraying (try common passwords against many accounts).

Q202 What is physical security and why does it matter?

Ans: Physical security prevents unauthorized physical access to systems. Measures: access control (key cards, biometrics, PIN), security cameras (CCTV), cable locks (laptops), server room physical locks, privacy screens, badge policies, visitor management, clean desk policy, and secure disposal of sensitive documents (shredding).

Q203 What is screensaver/lock screen security?

Ans: Setting an automatic screen lock (password-protected screensaver or lock screen) after idle time prevents unauthorized access when user is away. Best practice: 5-15 minute timeout in most environments, immediately lock when leaving (Windows+L shortcut). Configurable via Group Policy in enterprise environments.

Q204 What is a security token?

Ans: A security token is a physical or software device generating one-time codes for MFA. Hardware: RSA SecurID key fob, YubiKey (FIDO2/U2F). Software: Google Authenticator, Microsoft Authenticator (TOTP — Time-Based One-Time Password). Hardware tokens are more phishing-resistant than SMS codes.

Q205 What is the difference between authentication and authorization?

Ans: Authentication (AuthN) verifies identity — who are you? Methods: password, biometrics, MFA. Authorization (AuthZ) determines what an authenticated identity can do — what are you allowed to access? Implemented via ACLs, RBAC (Role-Based Access Control), permissions. You authenticate first, then the system authorizes your actions.

Q206 What is RBAC (Role-Based Access Control)?

Ans: RBAC assigns permissions to roles (job functions) rather than individual users. Users are assigned to roles; they inherit the role's permissions. Benefits: simplified management (change role = change permissions for all members), principle of least privilege compliance, easier auditing. Common model in Active Directory and cloud IAM.

Q207 What is an acceptable use policy (AUP)?

Ans: AUP is a written policy users must agree to before accessing organizational IT resources. It defines: acceptable and prohibited activities (no personal use of company devices, no illegal activity), monitoring disclosure (employer may monitor), consequences for violations. Legal protection for organization and clear expectations for users.

Q208 What is a zero-day vulnerability?

Ans: A zero-day is a vulnerability unknown to the software vendor (or for which no patch exists) that can be exploited by attackers. High value because no defense exists yet. Mitigations: defense in depth (multiple security layers), behavioral detection (EDR), network segmentation to limit blast radius until patch is released.

Q209 What is BIOS/UEFI security and how do you harden it?

Ans: BIOS/UEFI hardening: set admin/user BIOS password (prevent unauthorized changes), enable Secure Boot (prevents unsigned bootloaders/OS), disable unnecessary boot devices (USB, optical — prevent bootable attack tools), enable TPM, set boot order to internal drive first, update firmware regularly for vulnerability patches.

Q210 What is certificate pinning?

Ans: Certificate pinning (HPKP) hardcodes expected certificate or public key information for a specific domain into an application. Prevents MITM attacks using rogue but CA-signed certificates. Used in mobile banking apps, VPN clients. If pinned certificate expires/changes without app update, connection fails — requires careful management.

SECTION 08

Software Troubleshooting

Malware Removal, OS Issues, Application Problems, Mobile Support

Q211 – Q240

Q211 What is the CompTIA best practice procedure for malware removal?

Ans: 8-step process: 1) Investigate/identify symptoms, 2) Quarantine infected system (disconnect network), 3) Disable System Restore (prevents malware hiding in restore points), 4) Remediate – update antivirus definitions and run full scan, 5) Schedule scans + run updates, 6) Enable System Restore + create restore point, 7) Educate end user, 8) Document findings.

TIP *This 8-step malware removal process is directly tested on Core 2!*

Q212 What are common malware infection symptoms?

Ans: Malware symptoms: antivirus disabled or won't update, unusual high CPU/disk/network activity, unexpected pop-up ads (adware/spyware), browser homepage changed without permission, new toolbars or extensions, PC performance degraded, files encrypted with ransom note, missing files, unexpected outbound network connections, rogue security software warnings.

Q213 How do you troubleshoot Windows that won't start?

Ans: Options: Startup Repair (automatic repair via Windows Recovery Environment), Safe Mode (minimal boot for diagnosis), System Restore (roll back to working state), Command Prompt in WRE (bootrec /fixmbr, bootrec /fixboot, bootrec /rebuildbcd), Startup Settings (disable driver enforcement), factory reset (last resort), or clean install.

Q214 What is Windows Recovery Environment (WRE) and how do you access it?

Ans: WRE provides recovery tools when Windows won't boot normally. Access: hold Shift while clicking Restart, boot from Windows installation media, or automatically appears after multiple failed boots. Tools: Startup Repair, System Restore, System Image Recovery, Command Prompt, Uninstall Updates, Reset this PC.

Q215 What is the difference between 'Reset this PC' and a clean install?

Ans: Reset this PC (Keep my files): removes apps and settings but keeps personal files – quick option. Reset this PC (Remove everything): removes all files, apps, and settings – like a clean install but without media. Clean install from bootable media: complete fresh start, ensures all traces of previous installation are removed, recommended for persistent issues.

Q216 How do you remove a browser hijacker?

Ans: Steps: run anti-malware scan (Malwarebytes), uninstall suspicious browser extensions (all browsers), check and remove suspicious programs in Programs and Features, reset browser to defaults (settings □ reset), check DNS settings (malware may redirect to rogue DNS), clear browser cache/cookies, verify hosts file (C:\Windows\System32\drivers\etc\hosts) for unauthorized entries.

Q217 What is application compatibility mode in Windows?

Ans: Compatibility mode allows older applications designed for earlier Windows versions to run on current Windows. Right-click executable □ Properties □ Compatibility tab □ Run this program in compatibility mode for [Windows version]. Also run as administrator, disable full-screen optimizations, change DPI scaling. Used for legacy business applications.

Q218 What causes the 'Application has stopped working' error?

Ans: Causes: corrupted application files (reinstall), missing required DLL or Visual C++ redistributable (install correct version), incompatible Windows version (compatibility mode), insufficient permissions (run as administrator), corrupted user profile (create new profile), resource conflict, or application bug (check for updates/patches from vendor).

Q219 What is a user profile corruption and how do you fix it?

Ans: Corrupt profile symptoms: settings reset on login, empty desktop, applications won't start properly, missing personal customizations. Fix: create new local user account, copy files from old profile (C:\Users\[old]) to new, or modify registry (HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList) to point to backup profile copy.

Q220 How do you troubleshoot printing issues on Windows?

Ans: Checklist: physical connections and power, correct default printer selected, clear print queue (services.msc □ Print Spooler □ restart service or manually delete files in C:\Windows\System32\spool\PRINTERS), update/reinstall printer driver (from manufacturer website), check Windows compatibility (32-bit printer driver issue on 64-bit), share permissions for network printer.

Q221 What is a DLL and what causes DLL errors?

Ans: DLL (Dynamic Link Library) contains shared code/resources used by multiple programs simultaneously. DLL errors: 'DLL not found' (missing file — reinstall application or install required runtime like Visual C++), DLL hell (conflicting versions from different apps — side-by-side assemblies resolve this), corrupt DLL (reinstall/SFC scan).

Q222 What is the Windows Event Viewer used for in software troubleshooting?

Ans: Event Viewer (eventvwr.msc) logs: Application errors (crash details, Event IDs for specific errors), Security events (failed logins), System events (driver failures, service crashes). For software issues: filter Application/System logs by Error level, note Event IDs, look for patterns before problem time, search Event IDs online for solutions.

Q223 What is a startup program and how do you manage them?

Ans: Startup programs run automatically at Windows login, consuming resources. Disable unnecessary ones: Task Manager □ Startup tab (right-click □ Disable), or msconfig □ Startup tab. Some programs add to registry: HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run (user), HKLM same path (all users). Malware often uses Run keys for persistence.

Q224 What is the Windows hosts file?

Ans: The hosts file (C:\Windows\System32\drivers\etc\hosts) maps hostnames to IP addresses before DNS is queried. Can be used to: block websites (map domain to 127.0.0.1), override DNS for testing, set up local name resolution. Malware modifies hosts file to redirect legitimate sites to malicious IPs. Check for unauthorized entries.

Q225 What is a corrupt Windows profile and how do you identify it?

Ans: Signs: temporary profile loaded at login ('You have been logged on with a temporary profile'), settings don't persist, Event ID 1500/1511 in Application log, desktop is empty, user-specific settings lost after each logout. Caused by: profile disk full, permission issues, power failure during profile save.

Q226 What are common Android troubleshooting steps?

Ans: Clear app cache/data (Settings ▢ Apps ▢ specific app), force stop misbehaving app, check storage (insufficient storage causes many issues), check battery optimization settings, update OS and apps, factory reset (last resort — backup first), safe mode (disables third-party apps to isolate issue, hold Power ▢ long-press Power off ▢ Safe Mode).

Q227 What are common iOS troubleshooting steps?

Ans: Force close problematic app, clear Safari cache, check iCloud storage, restart device (force restart: Volume Up ▢ Volume Down ▢ hold Power), update iOS and apps, reset settings (preserves data, resets all settings to defaults), erase all content and settings (complete factory reset — backup via iCloud/iTunes first).

Q228 What is app sideloading and its risks?

Ans: Sideloading installs applications from outside the official app store (APK files on Android, IPA on iOS via developer tools). Risks: malware-infected apps not vetted by app store security review, no automatic updates for security patches, voids warranty, may install stalkerware. Required for some legitimate enterprise deployments.

Q229 What is a frozen/unresponsive application and how do you handle it?

Ans: An unresponsive application has stopped processing UI events. On Windows: Task Manager ▢ right-click process ▢ End Task (or End Process Tree for all child processes). If Task Manager is unresponsive too: Ctrl+Alt+Delete ▢ Task Manager, or open Command Prompt/PowerShell and use 'taskkill /f /pid [PID]'.

Q230 What is Windows activation and what happens when it fails?

Ans: Windows activation verifies the license is valid and not used on more machines than permitted. Without activation: watermark displayed, personalization disabled (wallpaper/themes), nag notifications. Activation failure: hardware change triggers re-activation (call Microsoft or use Activation Troubleshooter), OEM license tied to motherboard, volume licenses use KMS server.

Q231 What is a missing operating system error?

Ans: Causes: corrupted or missing boot files (BCD, bootmgr), incorrect BIOS boot order, MBR corruption, failed storage drive, intentional dual-boot misconfiguration. Fix: boot from Windows installation media □ Repair your computer □ Troubleshoot □ Advanced Options □ Command Prompt □ run bootrec /fixmbr, /fixboot, /scanos, /rebuildbcd.

Q232 What is the difference between a virus and a worm in terms of behavior?

Ans: Virus attaches to a legitimate host file and activates when the infected file is executed — requires user action to spread. Worm is self-replicating, spreading automatically across networks by exploiting vulnerabilities — no user action needed. Worms spread much faster and can cause network congestion through self-propagation.

Q233 What is a trojan and how does it differ from a virus?

Ans: A trojan disguises itself as legitimate/useful software to trick users into installing it. Unlike viruses, trojans do not self-replicate. Once installed, they may: create backdoors, download additional malware (dropper), steal credentials (RAT — Remote Access Trojan), encrypt files (ransomware component), join a botnet.

Q234 What is a botnet?

Ans: A botnet is a network of malware-infected computers (bots/zombies) controlled remotely by a command-and-control (C2) server operated by an attacker. Used for: DDoS attacks, spam campaigns, credential stuffing, cryptocurrency mining, distributed password cracking. Individual bot owners are often unaware their machine is infected.

Q235 How do you troubleshoot slow internet on a PC when other devices are fine?

Ans: PC-specific: update/reinstall NIC driver, run network troubleshooter, check DNS settings (try 8.8.8.8), flush DNS cache (ipconfig /flushdns), reset TCP/IP stack (netsh int ip reset), reset Winsock (netsh winsock reset + reboot), check for malware consuming bandwidth (Task Manager ▢ Network column), disable unused network adapters.

Q236 What is the Windows Firewall with Advanced Security?

Ans: wf.msc provides advanced firewall management: create custom inbound/outbound rules per program/port/protocol/IP, monitor active connections, configure connection security rules (IPsec), set firewall profiles (Domain/Private/Public), import/export rules. More granular than basic Windows Firewall in Control Panel.

Q237 What is application allow/deny listing?

Ans: Application allowlisting (whitelisting) permits only approved applications to run – more secure but high management overhead. Application denylisting (blacklisting) blocks known bad applications – easier to manage but reactive. Allowlisting is recommended by NIST and DoD for high-security environments. Windows: AppLocker (Enterprise), Software Restriction Policies.

Q238 What does it mean if an application needs 'Run as Administrator'?

Ans: Some applications require elevated privileges to: write to protected registry locations, modify system files, install drivers, or access hardware directly. Right-click ▢ Run as administrator, or set Always run as administrator in Properties ▢ Compatibility. Risk: running all apps as admin removes UAC protection – only use when necessary.

Q239 How do you recover files from a PC that won't boot?

Ans: Options: boot from Linux live USB and access NTFS drive (majority of data accessible), use Windows WRE Command Prompt to copy files to USB, remove drive and connect as secondary to working PC, use data recovery software (Recuva, Stellar, TestDisk) for corrupted file systems, professional recovery service for physically damaged drives.

Q240 What is the best practice for software updates?

Ans: Enable automatic security updates for OS and applications. Test updates in non-production environment before enterprise-wide deployment. Use WSUS (Windows Server Update Services) for centralized Windows update management. Track CVE (Common Vulnerabilities and Exposures) advisories for critical vulnerabilities. Patch within 14-30 days of release for critical patches.

SECTION 09

Operational Procedures

Safety, Documentation, Professionalism, Environmental, Change Management

Q241 – Q270

Q241 What are best practices for IT workplace safety?

Ans: Electrical safety: disconnect power before working inside PC, use anti-static wrist strap, never work on carpeted floor with components, be aware of capacitor charge in CRT/PSU (discharge before touching). Chemical safety: read SDS (Safety Data Sheets) for cleaning chemicals, proper ventilation. Lifting safety: bend knees for heavy equipment, get help for server racks.

TIP *Safety is tested in Core 2 Operational Procedures – don't skip this domain!*

Q242 What is ESD and how do you prevent it?

Ans: ESD (Electrostatic Discharge) is the sudden flow of static electricity between objects at different potentials – can damage or destroy electronics with as little as 10V (humans feel discharge at ~3,000V). Prevention: anti-static wrist strap (grounded to chassis), ESD mat, anti-static bags for storage, avoid carpeted surfaces, handle PCBs by edges.

Q243 What is the purpose of a Material Safety Data Sheet (SDS/MSDS)?

Ans: SDS (Safety Data Sheet, formerly MSDS) provides: chemical product information, hazard identification, composition, first-aid measures, fire-fighting, handling/storage, exposure controls/PPE, physical/chemical properties, toxicological information, regulatory information. Required by OSHA for all hazardous materials in workplace. Must be accessible to all employees.

Q244 What is proper disposal of toner cartridges and batteries?

Ans: Toner cartridges: return to manufacturer recycling programs or certified e-waste recycler – never in regular trash (fine powder environmental hazard). Batteries: lithium-ion/polymer (laptops, phones) must not go in regular trash – take to electronics retailer recycling or hazardous waste facility. Alkaline batteries: check local regulations – many areas allow regular disposal.

Q245 What is change management in IT?

Ans: Change management is a formal process ensuring IT changes are properly planned, tested, approved, implemented, and documented to minimize disruption. Key steps: change request, risk assessment, change approval (CAB – Change Advisory Board), scheduling (change window), implementation, testing, rollback plan, post-implementation review, documentation.

Q246 What is a change advisory board (CAB)?

Ans: CAB is a group of stakeholders who review, evaluate, and approve or reject proposed IT changes. Typically includes: IT managers, system owners, security team, business representatives. Emergency CAB (ECAB) handles urgent changes requiring immediate action. CAB ensures changes are documented, risk-assessed, and communicated.

Q247 What is an incident response plan?

Ans: An incident response plan defines procedures for detecting, containing, and recovering from security incidents. Steps: Preparation □ Identification □ Containment □ Eradication □ Recovery □ Lessons Learned (PICERL). Minimize damage and recovery time while preserving evidence. Must be documented, tested (tabletop exercises), and regularly updated.

Q248 What is documentation and what types matter in IT?

Ans: Critical IT documentation: network diagrams (physical and logical), asset inventory (hardware/software), incident reports, change logs, standard operating procedures (SOPs), knowledge base articles (how-to guides), runbooks (step-by-step operational procedures), cable/port labeling, vendor contacts, and licensing records.

Q249 What is a ticketing system and its components?

Ans: IT ticketing systems track service requests and incidents. Components: ticket creation (user reports issue), categorization and priority assignment (P1 critical □ P4 low), assignment to technician, SLA tracking (response/resolution time), escalation (if SLA breached), knowledge base, reporting/analytics, and closure documentation. Examples: ServiceNow, Jira, Zendesk.

Q250 What is asset management?

Ans: Asset management tracks organizational IT assets through lifecycle: procurement, deployment, maintenance, and disposal. Includes: hardware inventory (serial numbers, location, user), software licenses (compliance monitoring), warranty tracking, and ITAD (IT Asset Disposition). Tools: Lansweeper, ManageEngine, ServiceNow ITAM. Required for compliance audits.

Q251 What is a knowledge base in IT support?

Ans: A knowledge base is a searchable repository of documented solutions, how-to guides, troubleshooting procedures, and FAQs. Benefits: self-service (users solve own issues), consistency (same solution applied by all technicians), faster resolution, onboarding new staff, and reducing repeat tickets. Updated regularly with new solutions.

Q252 What is professional communication in IT support?

Ans: Key communication practices: active listening (let user finish without interrupting), use non-technical language with end users, document conversations, follow up to confirm issue resolution, maintain positive attitude, avoid blame/negativity, respect user's time, escalate appropriately, and set realistic expectations for resolution timelines.

Q253 What is a service level agreement (SLA) in IT support?

Ans: SLA defines committed service levels between IT and customers: response time (time to acknowledge ticket), resolution time (time to resolve by priority), uptime guarantees, support hours, escalation paths. Breach procedures: automatic escalation, management notification, compensation. SLAs drive IT team priorities.

Q254 What is remote monitoring and management (RMM)?

Ans: RMM software allows MSPs (Managed Service Providers) and IT teams to remotely monitor and manage endpoints, servers, and networks. Features: real-time alerts, remote access, patch management, script deployment, performance dashboards, automated remediation. Examples: NinjaRMM, ConnectWise Automate, N-able.

Q255 What is a business continuity plan (BCP)?

Ans: BCP ensures critical business functions continue during/after disasters: natural disasters, cyberattacks, power outages, hardware failures. IT BCP includes: DR plan (data center recovery), backup procedures, failover systems, communication plan, emergency contacts, alternate work locations, and regular testing (tabletop exercises, disaster drills).

Q256 What is the 3-2-1 backup rule?

Ans: 3-2-1 rule: keep 3 copies of data (1 primary + 2 backups), on 2 different media types (e.g., local NAS + tape or USB), with 1 copy stored offsite (cloud, remote facility). Protects against: hardware failure (2nd copy), local disaster (offsite copy), media failure (2 media types). Modern variant: 3-2-1-1-0 (1 offline, 0 errors verified).

Q257 What is the difference between RPO and RTO?

Ans: RPO (Recovery Point Objective): maximum acceptable data loss measured in time – 'how much data can we afford to lose?' Drives backup frequency. RTO (Recovery Time Objective): maximum acceptable downtime – 'how quickly must we restore?' Drives DR system investment. Shorter RPO/RTO = higher cost. Critical systems: RPO minutes/RTO hours.

Q258 What is environmental impact in IT operations?

Ans: IT environmental considerations: power consumption (PUE – Power Usage Effectiveness, target <1.5), cooling efficiency, e-waste recycling, virtualization to reduce physical servers, energy star-rated equipment, power management settings (sleep/hibernate), green procurement policies, and carbon footprint reporting. Data centers consume ~1-2% of global electricity.

Q259 What is proper workstation ergonomics?

Ans: Ergonomic setup: monitor at eye level (top of screen at or slightly below eye level), arm's length distance, keyboard at elbow height (90 degree arms), wrists neutral (not bent), feet flat on floor or footrest, chair supports lower back, take 20-20-20 breaks (every 20 min, look 20 feet away for 20 sec), reduce glare on monitor.

Q260 What is regulated data and how must it be handled?

Ans: Regulated data types: PII (Personally Identifiable Information), PHI (Protected Health Information — HIPAA), PCI data (payment cards — PCI-DSS), FERPA (student records), GDPR (EU personal data), government classified data. Handling requirements: encryption, access controls, audit logging, retention policies, secure disposal, and breach notification obligations.

Q261 What is the chain of custody in IT forensics?

Ans: Chain of custody documents the collection, transfer, storage, and analysis of evidence for legal proceedings. Each handler signs for the evidence; documentation records who had it, when, and what was done to it. Maintaining chain of custody ensures evidence is admissible in court. IT technicians may need to preserve evidence after an incident.

Q262 What are prohibited content and licensing violations?

Ans: Prohibited activities: installing unlicensed software (software piracy — violation of EULA and copyright law), distributing copyrighted content, accessing CSAM, storing regulated data without authorization. Consequences: DMCA notices, lawsuits, criminal charges. IT policies must prohibit these activities and monitoring must detect violations.

Q263 What is a non-disclosure agreement (NDA) in IT?

Ans: NDA is a legal contract requiring parties to keep specified information confidential. IT technicians often sign NDAs because they: access sensitive systems, see confidential business data, handle customer PII. Violation consequences: civil liability, termination. Read NDAs carefully before signing — they may have very broad scope.

Q264 What is PII and why must it be protected?

Ans: PII (Personally Identifiable Information) is data that can identify a specific individual: name + SSN, email address, phone number, date of birth, fingerprints, IP address, device ID, health data, financial data. Regulations require: encryption at rest and in transit, access logging, minimum necessary collection, breach notification.

Q265 What is the role of IT in disaster recovery?

Ans: IT DR responsibilities: identify critical systems (via BIA), establish backup and replication strategies, document recovery procedures (runbooks), test recovery procedures regularly (DR drills), maintain vendor contacts for emergency support, configure monitoring/alerting for early detection, and ensure documented failover/failback procedures.

Q266 What is a site survey for wireless networks?

Ans: Wireless site survey maps RF coverage, identifies sources of interference (microwaves, other Wi-Fi networks, Bluetooth, walls/floors), determines optimal AP placement for coverage and capacity. Types: passive survey (listen only), active survey (connect and test throughput), predictive survey (software simulation before installation).

Q267 What is the importance of labeling in IT infrastructure?

Ans: Consistent labeling: network cables (both ends with circuit ID), patch panel ports, rack equipment (asset tag, hostname, IP), power strips (which circuit), server ports. Benefits: faster troubleshooting (immediately identify cables), accurate documentation, safer changes (avoid removing wrong cable), compliance requirements.

Q268 What is a help desk call handling best practice?

Ans: Best practices: greet professionally, verify caller identity before accessing accounts, active listening, document ticket while on call, set realistic expectations, don't promise what you cannot deliver, follow escalation procedures, never access more data than needed to resolve the issue, follow up to confirm resolution, document solution for knowledge base.

Q269 What is onboarding and offboarding from an IT perspective?

Ans: IT onboarding: provision accounts (AD, email, VPN), configure workstation, install required software, provide access badges, security awareness training. IT offboarding: immediately disable all accounts (same-day for terminated employees), revoke badges/physical access, retrieve devices, transfer data per retention policy, audit for data exfiltration before offboarding.

TIP *Account offboarding on the same day of termination is a security critical process!*

Q270 What is a network/service monitoring tool?

Ans: Monitoring tools track system/network health and alert on issues. Types: Infrastructure monitoring (Nagios, PRTG, Zabbix – CPU, memory, disk, network interfaces), Application Performance Monitoring (APM – New Relic, Datadog), Log management (Splunk, ELK Stack), Network traffic analysis (Wireshark, SolarWinds NTA). Enable proactive issue resolution.

SECTION 10

Career, Certification & Advanced Topics

A+ Career Paths, Core 1+2 Summary, CompTIA Roadmap, Advanced Concepts

Q271 – Q300

Q271 What are the two CompTIA A+ exams and their domains?

Ans: Core 1 (220-1101) domains: Mobile Devices 15%, Networking 20%, Hardware 25%, Virtualization and Cloud 11%, Hardware and Network Troubleshooting 29%. Core 2 (220-1102) domains: Operating Systems 31%, Security 24%, Software Troubleshooting 26%, Operational Procedures 19%. Both exams required for A+ certification.

TIP *Know the exact domain percentages – they tell you what to prioritize studying!*

Q272 What is the passing score for CompTIA A+ exams?

Ans: Both Core 1 (220-1101) and Core 2 (220-1102) require a score of 675/900 to pass. Exams have up to 90 questions, including multiple choice, drag-and-drop, and performance-based questions (PBQs). PBQs are scenario simulations – more complex but worth more points. Total exam time: 90 minutes per exam.

Q273 What are performance-based questions (PBQs) on CompTIA exams?

Ans: PBQs are scenario-based simulations requiring hands-on application of knowledge: drag-and-drop network diagrams, configure virtual systems, match troubleshooting steps to scenarios, identify components in diagrams. They appear early in exams and take longer – don't get stuck; mark and return to PBQs if needed.

Q274 What IT career paths does CompTIA A+ open?

Ans: A+ certifies for: Help Desk Technician (Tier 1/2 support), Desktop Support Technician, IT Support Specialist, Field Service Technician, NOC Technician. Salary range (India): ₹3-6 LPA entry level; ₹6-12 LPA with 3-5 years experience. US: \$40K-\$60K entry, \$55K-\$85K with experience.

TIP *A+ + Network+ + Security+ is the classic CompTIA progression for IT careers.*

Q275 What is the CompTIA certification roadmap?

Ans: Foundation: ITF+. Core: A+ (IT support), Network+ (networking), Security+ (cybersecurity — DoD Directive 8570 baseline). Infrastructure: Server+, Cloud+. Cybersecurity: CySA+ (analyst), PenTest+ (ethical hacking), CASP+ (advanced security — manager/architect level). Professional: Project+. Each builds on previous skills.

Q276 What is CompTIA CertMaster and how does it help A+ candidates?

Ans: CertMaster is CompTIA's official learning platform offering: CertMaster Learn (full interactive course), CertMaster Practice (adaptive Q&A; engine with performance analytics), CertMaster Labs (virtual browser-based hands-on labs). Subscription-based. Complements but doesn't replace hands-on lab practice with real hardware.

Q277 What is an MSP (Managed Service Provider)?

Ans: MSP provides outsourced IT management to businesses on a subscription/flat-fee model. Services: remote monitoring, patch management, helpdesk, backup management, security services, network management. A+ certified technicians are commonly employed by MSPs. Key MSP tools: RMM, PSA (Professional Services Automation), documentation platforms.

Q278 What is ITIL and how does it relate to A+ skills?

Ans: ITIL (IT Infrastructure Library) v4 provides ITSM (IT Service Management) best practices. A+ aligns with ITIL concepts: incident management (troubleshooting), change management (change procedures), asset management, knowledge management (documentation), and service desk operations. ITIL Foundation certification complements A+ for service management roles.

Q279 What is the difference between a break/fix technician and an MSP model?

Ans: Break/fix: called when something breaks, charges per incident/hour — reactive model, no ongoing relationship, customer pays only when broken. MSP: proactive monitoring and maintenance on subscription — prevents issues before they occur, predictable monthly cost for customer, recurring revenue for MSP. Industry trend is toward MSP model.

Q280 What is CompTIA's continuing education (CE) program?

Ans: CompTIA certifications are valid for 3 years. Renewal via: earning CEUs (Continuing Education Units) through training, conferences, and higher-level certifications, or retaking the current exam. Reporting via CompTIA's CertMaster CE portal. A+ renewal: 20 CEUs required. Higher-level certifications (e.g., Network+) automatically renew A+.

Q281 What is a ticketing system's role in IT career development?

Ans: Working with ticketing systems builds: structured troubleshooting habits, documentation discipline, communication skills (written updates for users), SLA awareness, pattern recognition (recurring issues), knowledge base contribution, and metrics for performance evaluation (MTTR — Mean Time to Repair, CSAT — Customer Satisfaction scores).

Q282 What is the difference between Tier 1, 2, and 3 IT support?

Ans: Tier 1 (L1): First contact, handles simple issues — password resets, basic troubleshooting, ticket routing. Tier 2 (L2): More complex issues — advanced OS, network, hardware troubleshooting requiring deeper expertise. Tier 3 (L3): Specialist/engineering level — escalated complex issues, product specialists, developers, vendor engagement.

Q283 What is scripting and automation for A+ technicians?

Ans: Basic scripting knowledge improves efficiency: Batch scripts (.bat) for simple Windows automation, PowerShell for Windows administration (Active Directory, file management, software deployment), Bash for Linux, Python for cross-platform. Automate: repetitive tasks, user provisioning, software deployment, report generation.

Q284 What is the importance of vendor documentation and resources?

Ans: Vendor documentation: hardware service manuals (teardown procedures, FRU lists), firmware release notes (what bugs are fixed), technical whitepapers, knowledge base articles, driver downloads. Resources: vendor support portals, community forums, subreddits, YouTube channels. Always consult official docs before attempting repairs.

Q285 What is fiber optic cabling and when is it used?

Ans: Fiber optic transmits data as light pulses through glass/plastic fiber. Types: Single-mode (SMF, small core ~9 micron, long distance up to 100 km) for WAN/backbone; Multi-mode (MMF, large core ~50-62.5 micron, up to 550m) for LAN. Benefits: immune to EMI, very high bandwidth, low attenuation. Connectors: SC, LC, ST, MTP/MPO.

Q286 What is a PBX and VoIP phone system?

Ans: PBX (Private Branch Exchange) is an internal telephone switching system connecting internal phones to each other and to external phone lines. Modern IP-PBX (like Cisco Unified CM, 3CX, Asterisk) uses VoIP. A+ technicians may configure VoIP phones, troubleshoot QoS, and register phones to PBX. Use dedicated VLAN for voice traffic.

Q287 What are common cable management best practices?

Ans: Cable management: use cable trays/raceways to organize runs, label both ends of all cables, bundle with Velcro ties (not zip ties – too tight damages cables), maintain bend radius (Cat6 min 4x cable diameter), keep power and data cables separated (EMI), document cable plant (which port connects to what), use proper length cables.

Q288 What is a rack unit (U) in data center terminology?

Ans: A rack unit (U) is a standard unit of measurement for rack-mountable equipment height. 1U = 1.75 inches (44.45mm). Standard server rack is typically 42U tall. Servers are 1U, 2U, 4U; storage arrays may be 2-4U; network switches are typically 1-2U. Knowing U sizes is essential for rack planning and capacity management.

Q289 What is IPAM (IP Address Management)?

Ans: IPAM tracks and manages IP address allocation across a network. Features: IP address tracking (who has which IP), DNS and DHCP integration (DDI), subnet planning, conflict detection, reporting. Tools: SolarWinds IPAM, Infoblox, ManageEngine OpManager. Critical for large networks to prevent IP conflicts and maintain accurate inventory.

Q290 What is the CompTIA A+ exam format and question types?

Ans: Format: 90 questions maximum, 90 minutes, scored 100-900 (pass at 675). Question types: multiple choice (single best answer), multiple select (choose 2-3 correct), drag-and-drop, PBQs (performance-based simulation scenarios). No open-book. Pearson VUE testing centers or online proctored. Photo ID required.

Q291 What is UEFI Secure Boot and how does it work?

Ans: Secure Boot (part of UEFI specification) verifies that boot software is signed by a trusted key before executing. Prevents bootkit/rootkit malware from loading before the OS. The UEFI firmware maintains a database of trusted signing keys. Required for Windows 11. Disable only when installing Linux that doesn't support Secure Boot or for legacy hardware.

Q292 What is the difference between BIOS and UEFI?

Ans: Legacy BIOS: 16-bit code, max 2 TB drive support (MBR), no mouse support in setup, slower POST, no secure boot. UEFI: 32/64-bit, supports drives over 2 TB (GPT), graphical mouse-enabled setup, faster POST (compressed initialization), Secure Boot, network stack for PXE boot, Compatibility Support Module (CSM) for legacy OS support.

Q293 What is Microsoft 365 from an IT support perspective?

Ans: Microsoft 365 (formerly Office 365) is Microsoft's cloud subscription suite: Exchange Online (email), SharePoint Online (collaboration/intranet), Teams (communication), OneDrive (cloud storage), Office apps (Word, Excel, PowerPoint). IT manages via Microsoft 365 Admin Center: user accounts, licenses, security policies, compliance.

Q294 What is Active Directory and why is it important for A+ technicians?

Ans: Active Directory (AD) is Microsoft's directory service for centralized management of users, computers, groups, and policies in a Windows domain. A+ technicians need: join computers to domain, manage user accounts (enable/disable/reset passwords), understand Group Policy application, and navigate AD Users and Computers (ADUC) console.

TIP *AD skills are essential for any enterprise IT support role!*

Q295 What is LDAP and how does Active Directory use it?

Ans: LDAP (Lightweight Directory Access Protocol, port 389/636 for LDAPS) is the protocol used to query and modify directory services like Active Directory. Applications authenticate against AD via LDAP. LDAP queries locate objects: CN (Common Name), OU (Organizational Unit), DC (Domain Component). LDAPS uses TLS encryption.

Q296 What is a domain vs workgroup in Windows?

Ans: Workgroup: peer-to-peer, each PC manages its own accounts, max ~10-20 PCs practical, no centralized management, used in homes/small offices. Domain: centralized management via Active Directory DC (Domain Controller), single sign-on, Group Policy enforcement, centralized authentication, scales to thousands of computers. Required for enterprise environments.

Q297 What is remote desktop and its security considerations?

Ans: RDP provides remote GUI access to Windows systems. Security considerations: disable RDP when not needed (attack surface), use VPN before connecting, enable Network Level Authentication (NLA), use strong passwords + MFA, change default port 3389, restrict via Windows Firewall to specific IPs, use RDP gateway for external access.

Q298 What are the most important things to know about A+ Core 2 security?

Ans: Core 2 security focus: malware types and removal procedure (8 steps), social engineering attacks, physical security, authentication methods (MFA, biometrics, certificates), encryption (BitLocker, EFS, TLS), Windows security features (Defender, Firewall, UAC), SOHO security hardening, data destruction methods, and regulatory compliance basics.

Q299 What study strategy is best for passing CompTIA A+?

Ans: Proven approach: study Core 1 and Core 2 separately, use official CompTIA study guide + video course (Professor Messer is free and excellent), build a home lab (VirtualBox for OS practice, old hardware for teardowns), do practice exams until consistently 80%+ (Darril Gibson, Mike Meyers, ExamCompass), review wrong answers thoroughly, attempt PBQs in practice.

Q300 What are the best practices for a long-term IT career after A+?

Ans: Build on A+ immediately: get hands-on experience (even volunteer/intern), earn Network+ then Security+ within 1-2 years, specialize (cloud: AWS/Azure, security: CySA+/CISSP track, or infrastructure: CCNA/RHCE), build a home lab, contribute to IT communities (Reddit r/sysadmin, Spiceworks), document everything (GitHub, blog), and never stop learning.

TIP *The A+ is just the beginning – compound your skills every year!*